The Simplest Derisking Model?
Drawdown Feedback, Daily Calibration, and Recovery

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September 8, 2026

Abstract

Firms adjust exposure to a sigmoid of drawdown in the systematic crowding index, with linear impact and geometric O-W-style recovery. A daily calibration reduces normal exposure 0.5 toward a floor of 0.3 across a 5–8% price-drawdown band. A sale of 0.05 moves the price by 30 basis points, and impact has a three-day half-life. The maximum feedback gain is 0.55, below the local instability boundary. With assumed daily fundamental volatility 0.5% and positive mean 3.125 basis points, independent simulations give mean calm spells of 79 trading days and 107 days between activations. An activated episode averages 28 days, split into 13 days to its minimum exposure and 15 days to recovery. Only 18% of completed episodes reach $B \le 0.31$; those take about 15 days to reach that level and 103 further days to recover. Mean and volatility strongly affect these times, while nearby half-lives have a smaller effect. The model retains generic $\mu > 0$; the numerical mean and volatility are assumptions rather than empirical estimates. Earlier normalized high-impact comparisons are preserved separately.

1 Mechanism

Hedge funds can hold different portfolios while sharing the same crowded trade. Consider N funds, and let $\mathbf{w}_i(t)$ be fund i's vector of signed dollar positions, expressed in fixed common exposure units. The single-factor representation is

$$\mathbf{w}_i(t) = b_i(t)\mathbf{u} + \mathbf{v}_i(t), \qquad b_i(t) > 0, \qquad \|\mathbf{u}\|_1 = 1. \tag{1.1}$$

The fixed dimensionless vector $\mathbf{u}$ describes the common trade, which can contain both long and short positions. Its normalization fixes the scale of the positive loading $b_i(t)$; this loading measures the fund's common exposure, rather than its total leverage. The residual $\mathbf{v}_i(t)$ contains its other holdings. We retain the assumption $E[\mathbf{v}_i(t)'\mathbf{v}_j(t)] = 0$ for $i \neq j$. This orthogonality concerns residual holdings; it does not by itself imply uncorrelated residual profits under a general asset covariance matrix.

Let $\mathbf{r}(t)$ be the vector of excess asset returns. We assume the factor model

$$\mathbf{r}(t) = \mathbf{B}\mathbf{f}(t) + \boldsymbol{\epsilon}(t), \tag{1.2}$$

where the fixed loading matrix $\mathbf{B}$ has one row per asset and one column per factor, $\mathbf{f}(t)$ contains factor returns, and $\boldsymbol{\epsilon}(t)$ contains asset-specific residual returns. The matrix $\mathbf{B}$ is distinct from aggregate exposure B_t. Define the systematic crowding return and projected asset residual by

$$r_c(t) := \mathbf{u}'\mathbf{B}\mathbf{f}(t), \qquad \epsilon_c(t) := \mathbf{u}'\boldsymbol{\epsilon}(t), \tag{1.3}$$

so the total return in the crowding direction is $\mathbf{u}'\mathbf{r}(t) = r_c(t) + \epsilon_c(t)$. For positions held fixed over a return interval, excess marked P&L decomposes as

$$\mathbf{w}_i(t)'\mathbf{r}(t+1) = b_i(t)r_c(t+1) + [b_i(t)\epsilon_c(t+1) + \mathbf{v}_i(t)'\mathbf{r}(t+1)]. \tag{1.4}$$

Thus a negative crowding return contributes a loss to every positively exposed fund, even when their residual holdings differ. Traders monitor drawdown in the systematic component r_c; the bracketed remainder affects fund profits but does not enter the exposure rule. Below, r_c drives a synthetic log crowding index. Its correspondence to compounded simple returns uses the usual small-return approximation. Aggregate crowding exposure is

$$B_t := \sum_{i=1}^{N} b_i(t), \qquad \sum_{i=1}^{N} \mathbf{w}_i(t) = B_t\mathbf{u} + \sum_{i=1}^{N} \mathbf{v}_i(t). \tag{1.5}$$

Common deleveraging reduces these loadings together. The scalar impact model below retains the price effect of changes in B_t and excludes impact from residual-position trading as a separate modeling assumption.

The intended behavior has three parts. Firms maintain their normal exposure through ordinary fluctuations. A sufficiently large drawdown induces sales; their price impact deepens the drawdown and induces further sales. Exposure eventually rebuilds as positive fundamental returns repair the loss. When impact is transient, the decay of earlier selling pressure also supports recovery. The symmetry matters: recovery of earlier buying pressure can then induce another drawdown. We compare this mechanism with permanent impact.

A sigmoid of a trailing return average does not isolate this mechanism: an old loss leaving the window can increase desired exposure even if prices have not recovered. We therefore replace the previous 40-period average with drawdown from the running price peak. This memory does not expire after 40 periods. The exposure target is flat in normal conditions and steep only near the loss threshold. Firms adjust fully to the current target in each period. Differences between liquidation and recovery arise from the feedback and the fundamental returns. There is no externally imposed regime switch or recovery reset.

2 Model and First-Order Dynamics

2.1 Exposure, Returns, and Drawdown

Time is discrete. Aggregate crowding exposure B_t is measured in fixed exposure units, with normal level $b_{\max} > 0$ and residual floor $0 < b_{\min} < b_{\max}$. The daily calibration uses $b_{\max} = 0.5$ and $b_{\min} = 0.3$; the earlier comparisons retain their normalization $b_{\max} = 1$. Individual exposures can be $b_i(t) = \bar{b}_i B_t / b_{\max}$, where $\bar{b}_i > 0$ is firm i's normal exposure and $\sum_{i=1}^{N} \bar{b}_i = b_{\max}$. This normalization ensures $\sum_i b_i(t) = B_t$, as in (1.5). Every firm remains positively exposed.

Use the systematic return $r_c(t+1)$ from (1.3) as the log increment of the crowding index, and let $\Delta B_{t+1} := B_{t+1} - B_t$ be the signed trade. Instantaneous impact is linear:

$$g : \mathbb{R} \longrightarrow \mathbb{R}, \qquad g(\Delta B) = \lambda \Delta B, \qquad \lambda > 0. \tag{2.1}$$

Introduce a signed price displacement I_t and let $\rho \in [0,1]$ be the fraction of past impact surviving one period. Write $\mathbf{f}^{\text{fund}}(t)$ for factor returns absent crowding impact and define

$$\xi_t := \mathbf{u}'\mathbf{B}\mathbf{f}^{\text{fund}}(t). \tag{2.2}$$

The fundamental systematic return ξ_t is distinct from the projected asset residual $\epsilon_c(t)$. We specify the law of ξ_t directly; no residual-return process is needed to simulate the exposure rule. The impact and return equations are

$$\begin{aligned} I_{t+1} &= \rho I_t + g(\Delta B_{t+1}), \\ r_c(t+1) &= \xi_{t+1} + I_{t+1} - I_t = \xi_{t+1} + \lambda \Delta B_{t+1} - (1-\rho) I_t, \\ \xi_{t+1} &= \mu + \sigma Z_{t+1}, \qquad Z_t \overset{\text{iid}}{\sim} N(0,1). \end{aligned} \tag{2.3}$$

Here $\mu := E[\xi_t] > 0$ and $\sigma := \mathrm{sd}(\xi_t) > 0$ are independent model parameters. The coefficient λ has units of log return per unit of exposure. Prices are observed after each period's trade; the new trade responds to the preceding observation, while old impact decays between trades. A negative displacement contributes a positive recovery return $-(1-\rho)I_t$; a positive displacement contributes a negative one.

This is a symmetric, log-price adaptation of the exponential resilience in Obizhaeva and Wang (2013, Section 3). Their block-shaped book gives a linear quote displacement and a quadratic total execution cost. We model the marked price used for drawdown, retain the prescribed exposure rule, and allow all impact to decay; we do not solve their optimal-execution problem or reproduce their separate bid and ask processes. The factor one-half in average execution impact is not a factor in the post-trade price displacement.

It is convenient to describe resilience by its half-life h, in periods:

$$\rho = 2^{-1/h}, \qquad h = 0 \Longleftrightarrow \rho = 0, \qquad h = \infty \Longleftrightarrow \rho = 1. \tag{2.4}$$

Thus larger h means slower recovery. At $\rho = 1$ and $I_0 = 0$, $I_t = \lambda(B_t - B_0)$ and the model reduces exactly to the permanent-impact benchmark. At $\rho = 0$, each trade's impact is fully reversed before the following trade, but the following trade creates fresh impact.

If p_t denotes the systematic crowding log index, define its drawdown by

$$d_t := \max_{s \le t} p_s - p_t \ge 0, \qquad d_{t+1} = \max\{0, d_t - r_c(t+1)\}. \tag{2.5}$$

A log drawdown d corresponds to a price drawdown $1 - e^{-d}$. An initial $d_0 > 0$ summarizes a price peak that predates the simulation.

2.2 A Saturating Target and Immediate Adjustment

Firms choose their next exposure from the current drawdown. They reach this target within one period, corresponding to $a = 1$ in the previous partial-adjustment rule. No adjustment-speed parameter is needed.

For a drawdown threshold $\theta > 0$ and transition width $w > 0$, set

$$L(d) = b_{\min} + \frac{A}{1 + \exp((d-\theta)/w)}, \qquad A := (b_{\max} - b_{\min})(1 + e^{-\theta/w}). \tag{2.6}$$

The normalization gives $L(0) = b_{\max}$ and $L(d) \downarrow b_{\min}$ as drawdown increases; A is determined by the other parameters. Small drawdowns leave desired exposure close to $b_{\max}$. Near θ, desired exposure falls sharply, and at large drawdowns it approaches the retained floor $b_{\min}$. A larger θ means firms tolerate more loss before deleveraging. A smaller w makes the response more abrupt. When θ/w is large, $L(\theta) \simeq (b_{\max} + b_{\min})/2$.

Exposure obeys

$$B_{t+1} = L(d_t). \tag{2.7}$$

The signed trade is simply the exposure change, $\Delta B_{t+1} = L(d_t) - B_t$. For example, if current exposure is 0.5 and the target is 0.4, firms sell 0.1 and reach exposure 0.4 in the next period. A fixed target would require no further trading. In the full system, however, this sale depresses the return and can lower the next target. Further sales then follow, even though each preceding target was reached in full. Figure 1 illustrates the daily calibration of the target and instantaneous impact.

The timing matters: B_{t+1} responds to d_t, before the new return and d_{t+1} are observed. The complete first-order state equation is

$$\begin{aligned} B_{t+1} &= L(d_t), \\ I_{t+1} &= \rho I_t + \lambda[L(d_t) - B_t], \\ d_{t+1} &= \max\{0, d_t - \xi_{t+1} - I_{t+1} + I_t\}\,. \end{aligned} \tag{2.8}$$

The initial conditions are $B_0 \in [b_{\min}, b_{\max}]$, $d_0 \geq 0$, and $I_0 = 0$. Since every target lies in $[b_{\min}, b_{\max}]$, this interval is invariant. Returns remain stochastic even when exposure is close to its ceiling. Saturation prevents unbounded leverage or negative positions; it does not remove the price feedback.

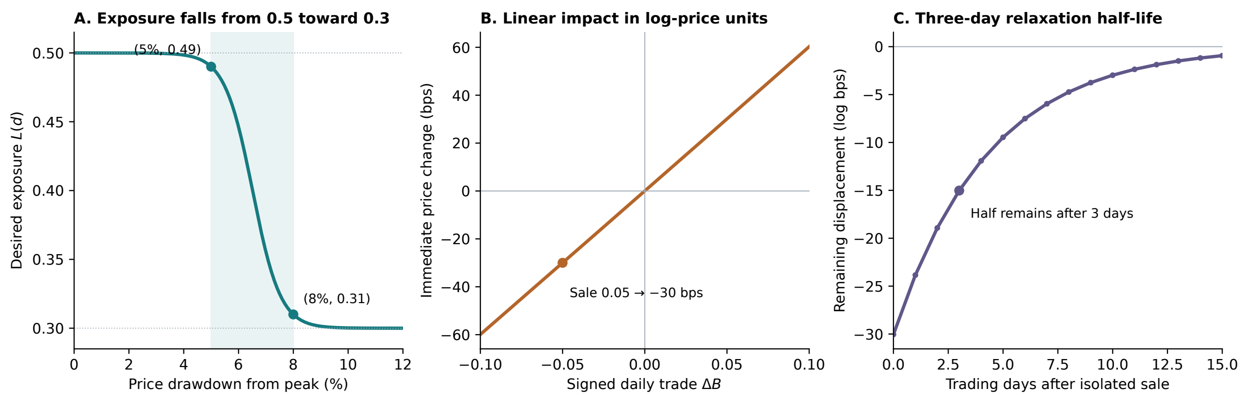

Figure 1: Daily calibration. Left: 5% and 95% of the exposure reduction occur at simple-price drawdowns of 5% and 8%, respectively; the target approaches its floor of 0.3. Center: the isolated price effect of a trade is linear in log-price units; a sale of 0.05 produces exactly −30 basis points in simple-price terms. Right: with no further trades, the sale's displacement halves every three trading days. The complete daily return also contains fundamental innovations and recovery of prior impact.

2.3 Dynamics and Notation at a Glance

The following equations collect the model in one place. Given the current state (B_t, d_t, I_t), traders choose their exposure adjustment before the next fundamental innovation is realized. The trade and the innovation determine the return, which updates drawdown:

$$
\begin{aligned}
A &= (b_{\max} - b_{\min})(1 + e^{-\theta/w}) && \text{Normalization} \\
L(d_t) &= b_{\min} + \frac{A}{1 + \exp((d_t - \theta)/w)} && \text{Desired exposure} \\
B_{t+1} &= L(d_t) && \text{Exposure adjustment} \\
g(\Delta B_{t+1}) &= \lambda \Delta B_{t+1} && \text{Instantaneous impact} \\
I_{t+1} &= \rho I_t + g(\Delta B_{t+1}) && \text{Impact memory} \\
\xi_{t+1} &= \mu + \sigma Z_{t+1} && \text{Fundamentals} \\
r_c(t+1) &= \xi_{t+1} + I_{t+1} - I_t && \text{Systematic return} \\
d_{t+1} &= \max\{0, d_t - r_c(t+1)\} && \text{Drawdown} \\
p_{t+1} &= p_t + r_c(t+1) && \text{Log price.}
\end{aligned}
\tag{2.9}
$$

Define the three-dimensional state vector $\mathbf{x}_t := (B_t, d_t, I_t)'$, observed after period t's trade. Its components are current exposure, log drawdown from the running price peak, and signed price displacement due to impact. Substituting the exposure and impact equations into the drawdown update gives the first-order stochastic difference equation system

$$
\mathbf{x}_{t+1} = \Phi(\mathbf{x}_t, Z_{t+1}) = \begin{pmatrix} L(d_t) \\ \max\{0,\, d_t - \mu - \sigma Z_{t+1} + (1-\rho) I_t - \lambda [L(d_t) - B_t]\} \\ \rho I_t + \lambda [L(d_t) - B_t] \end{pmatrix}. \tag{2.10}
$$

Here $Z_t \overset{\text{iid}}{\sim} N(0,1)$, with $0 < b_{\min} < b_{\max}$, $0 \le \rho \le 1$, and $\theta, w, \lambda, \mu, \sigma > 0$. Start from $\mathbf{x}_0 = (B_0, d_0, 0)'$, with $B_0 \in [b_{\min}, b_{\max}]$ and $d_0 \ge 0$. The system is first-order because the current state and the next independent innovation determine the next state. The drawdown component carries the relevant running-peak history. Returns and the log price are derived observables, not additional state variables: the return and log-price equations in (2.9) reconstruct them from any chosen p_0, with initial running peak $p_0 + d_0$. The mean μ remains an independent positive parameter; $\mu = \sigma/16$ is the baseline simulation setting used below.

Table 1 defines the symbols, including the derived quantities used in the analysis and the notation used only in illustrations or comparisons.

Ratios such as μ/σ, θ/σ, w/σ, and λ/σ express existing parameters in units of fundamental volatility; they introduce no additional parameters.

Table 1: Symbols and definitions. Exposure is measured in fixed units. The dynamic index is logarithmic; asset returns are excess simple returns.

Symbol	Definition
State, trades, and returns	
t, s	Discrete period indices; s also indexes the past price history.
N, i, j	Number of hedge funds and fund indices.
$\mathbf{w}_i(t)$, $\mathbf{u}$, $\mathbf{v}_i(t)$	Fund holdings in fixed dollar units, fixed common direction with $\|\mathbf{u}\|_1 = 1$, and residual holdings.
$b_i(t)$, $\bar{b}_i$	Positive individual and normal common exposures; $b_i(t) = \bar{b}_i B_t / b_{\max}$ and $\sum_i \bar{b}_i = b_{\max}$.
$\mathbf{r}(t)$	Excess asset-return vector.
$\mathbf{B}$, $\mathbf{f}(t)$	Fixed asset factor-loading matrix and factor-return vector.
$\mathbf{f}^{\text{fund}}(t)$	Factor returns absent crowding impact.
$\boldsymbol{\epsilon}(t)$, $\epsilon_c(t)$	Asset residual-return vector and its projection $\epsilon_c(t) = \mathbf{u}'\boldsymbol{\epsilon}(t)$.
$\mathbf{x}_t$, Φ	State vector $(B_t, d_t, I_t)'$ and transition map in (2.10).
B_t, B_0	Current and initial aggregate exposure, in $[b_{\min}, b_{\max}]$; normal exposure is $b_{\max}$.
d_t, d_0	Current and initial log drawdown; $d_t = \max_{s \le t} p_s - p_t \ge 0$.
ΔB_{t+1}	Signed exposure change $B_{t+1} - B_t$; positive for purchases and negative for sales.
I_t	Signed displacement of the marked log price from its unaffected value.
p_t	Systematic crowding log index; initial level p_0 is arbitrary.
$r_c(t+1)$	Systematic return $\mathbf{u}'\mathbf{B}\mathbf{f}(t+1)$, used as log-index increment $p_{t+1} - p_t$.
ξ_t, Z_t	Fundamental systematic log increment and its independent standard Gaussian innovation: $\xi_t = \mu + \sigma Z_t$.
Parameters and response functions	
$b_{\min}$, $b_{\max}$	Residual floor and normal exposure ceiling, with $0 < b_{\min} < b_{\max}$.
θ	Positive log-drawdown threshold at the sigmoid's inflection point.
w	Positive transition width, in log-drawdown units; smaller values make desired exposure fall more steeply.
λ	Positive linear impact coefficient, in log-return units per unit of exposure.
ρ, h	Surviving fraction of impact per period and its half-life; $\rho = 2^{-1/h}$, with endpoints zero and infinity.
μ, σ	Positive mean and standard deviation of ξ_t; independent parameters in log-return units.
$L(d)$	Desired exposure at log drawdown d, decreasing from $L(0) = b_{\max}$ toward $b_{\min}$.
A	Target normalization $(b_{\max} - b_{\min})(1 + e^{-\theta/w})$; determined by the other parameters.
$g(\Delta B)$	Signed market-impact function $g(\Delta B) = \lambda \Delta B$.
Derived quantities used in the analysis	
C_t, M_t	Coordinates $d_t + I_t$ and $B_t - I_t/\lambda$ used to separate fundamental returns from impact recovery.
δB_t, δI_t	Departures of exposure and impact from a fixed point; distinct from the trade ΔB_t.
$\mathbf{J}$, T, D	Jacobian of the reduced map, its trace $\rho + k$, and its determinant k.
k, z_1, z_2	Local gain $\lambda\|L'(C)\|$ at a fixed point, and the two eigenvalues of the reduced map; z denotes either eigenvalue.
G	Maximum feedback gain $\lambda A/(4w)$.
$L'(d)$	Derivative of desired exposure with respect to drawdown.
W_t, $F(d)$	Permanent-benchmark coordinate $d_t + \lambda B_t$ and equilibrium curve $d + \lambda L(d)$; used only when $\rho = 1$.
T_{peak}	First period when the price regains its initial running peak.
Illustrations and comparisons	
a	Fraction of the target gap traded in the earlier adjustment rule; fixed at one in the present model, and varied only in the comparison.

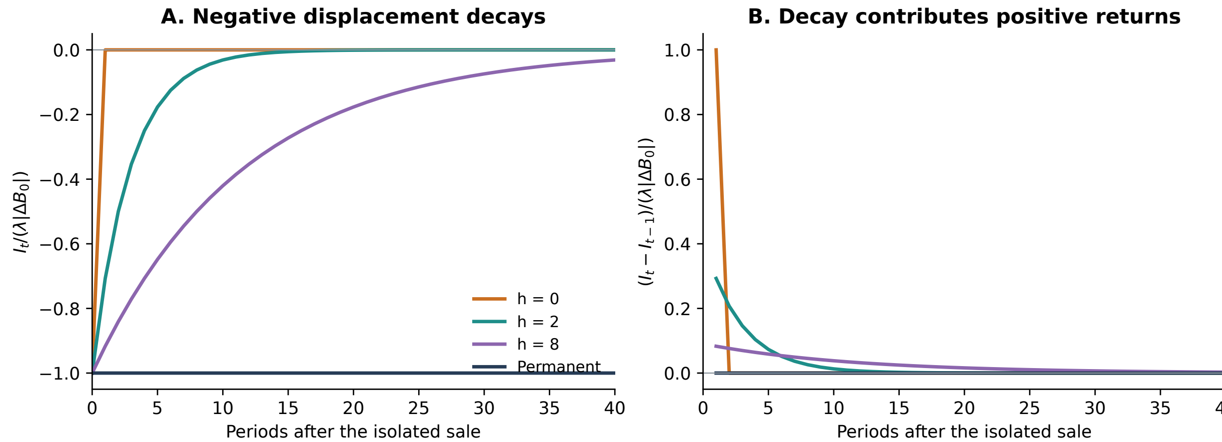

Figure 2: Impact recovery after an isolated sale. Fundamental increments and subsequent trades are zero for this illustration. Time zero is just after the sale, with displacement normalized to -1. Left: displacement follows $-\rho^t$. Right: its subsequent change contributes positive returns. The initial negative return is omitted from the right panel. Purchases give the same curves with signs reversed. Permanent impact has no recovery return.

3 Feedback, Resilience, and Recovery

3.1 Local Stability with Transient Impact

The goal is to determine whether the exposure–impact feedback makes a small disturbance die away or grow, and whether the adjustment oscillates. To isolate this mechanism, we set all subsequent fundamental increments to zero after an initial disturbance. This controlled calculation switches off both drift and noise; the full stochastic model retains $\mu > 0$. We first analyze a neighborhood in which the price remains strictly below its running peak, so the drawdown boundary does not bind.

The impact-adjusted drawdown $C_t = d_t + I_t$ satisfies the exact identity

$$C_{t+1} = \max\{I_{t+1}, C_t - \xi_{t+1}\}. \tag{3.1}$$

Below the peak, $d_{t+1} = d_t - \xi_{t+1} - I_{t+1} + I_t$, so $C_{t+1} = C_t - \xi_{t+1}$. With $\xi_{t+1} = 0$, C is constant: impact moves the price and drawdown, but their effects cancel in $d_t + I_t$. For a fixed $C > 0$, substitute $d_t = C - I_t$ to obtain the two-dimensional map

$$B_{t+1} = L(C - I_t), \qquad I_{t+1} = \rho I_t + \lambda[L(C - I_t) - B_t]. \tag{3.2}$$

At a fixed point, exposure no longer changes, so impact receives no new trades. For $0 \le \rho < 1$, the equality $I_* = \rho I_*$ then implies $I_* = 0$. Hence the unique fixed point at this value of C is $(B_*, d_*, I_*) = (L(C), C, 0)$. Different values of C give different fixed points. We study convergence within one fixed-C surface, rather than convergence of all initial states to one common exposure level.

Define the small departures $\delta B_t := B_t - L(C)$ and $\delta I_t := I_t$. These are deviations from equilibrium; δB_t is distinct from the one-period trade ΔB_t. Since $L'(C) < 0$, $L(C - I_t) \simeq L(C) + |L'(C)| I_t$. The linearized dynamics are therefore

$$\begin{pmatrix} \delta B_{t+1} \\ \delta I_{t+1} \end{pmatrix} \simeq \underbrace{\begin{pmatrix} 0 & |L'(C)| \\ -\lambda & \rho + k \end{pmatrix}}_{\mathbf{J}} \begin{pmatrix} \delta B_t \\ \delta I_t \end{pmatrix}, \qquad k := \lambda |L'(C)|. \tag{3.3}$$

The local gain k has a direct interpretation. An extra unit of drawdown reduces desired exposure by approximately $|L'(C)|$ units; the resulting sale adds k units of drawdown through instantaneous impact. Recovery of past impact enters separately through ρ.

For a linear discrete-time system, every small perturbation converges to zero precisely when both eigenvalues have modulus strictly less than one. Along an eigenmode, each step multiplies the perturbation by its eigenvalue; modulus below one gives decay, and modulus above one gives growth. The trace and determinant of $\mathbf{J}$ are $T := \operatorname{tr} \mathbf{J} = \rho + k$ and $D := \det \mathbf{J} = k$, so its eigenvalues solve

$$z^2 - Tz + D = z^2 - (\rho + k)z + k = 0. \tag{3.4}$$

The *strict Schur conditions* are the following necessary and sufficient inequalities for both roots of this real quadratic to lie inside the unit circle (Elaydi, 2007, Theorem 4.4):

$$1 - T + D > 0, \qquad 1 + T + D > 0, \qquad 1 - D > 0. \tag{3.5}$$

To see what they test, write the roots as z_1, z_2. The first two expressions are $(1 - z_1)(1 - z_2)$ and $(1 + z_1)(1 + z_2)$; the third requires $z_1 z_2 < 1$. If the roots are real, the first two inequalities put both inside $(-1, 1)$, both above 1, or both below -1. The product condition excludes the last two possibilities. If the roots are complex conjugates, their product is their common squared modulus, so the third inequality requires that modulus to be below one. Strict inequalities exclude eigenvalues on the unit circle, where linearization alone cannot decide nonlinear stability.

Substituting the model's trace and determinant gives

$$\begin{aligned} 1 - T + D &= 1 - \rho > 0 && \text{Some impact decays,} \\ 1 + T + D &= 1 + \rho + 2k > 0 && \text{Automatic for } \rho, k \geq 0, \\ 1 - D &= 1 - k > 0 && \text{Local feedback gain below one.} \end{aligned} \tag{3.6}$$

Thus, for $0 \leq \rho < 1$, the fixed point is locally asymptotically stable on its fixed-C surface when $k < 1$, and unstable when $k > 1$. At $k = 1$, both eigenvalues have unit modulus and nonlinear terms must be examined. The case $\rho = 1$ also fails the strict test and is treated separately in the next subsection.

Stability does not require a monotone return to equilibrium. The roots are complex when the discriminant $(\rho + k)^2 - 4k$ is negative, equivalently

$$(1 - \sqrt{1 - \rho})^2 < k < (1 + \sqrt{1 - \rho})^2, \tag{3.7}$$

and then have modulus $\sqrt{k}$. The oscillatory modes therefore decay for $k < 1$ and grow for $k > 1$, with amplitude multiplier $\sqrt{k}$ per period. Near $k = 1$, resilience therefore produces oscillatory feedback rather than an indefinitely one-way liquidation. Local growth can eventually leave the neighborhood where the approximation applies; it does not establish a nonlinear limit cycle.

To connect this local test to the parameter search, the maximum gain is

$$G := \max_{d \geq 0} \lambda |L'(d)| = \frac{\lambda A}{4w}. \tag{3.8}$$

If $G < 1$, every below-peak fixed point passes the strict test; if $G > 1$, some values of C have locally unstable feedback. The daily calibration has $G \simeq 0.55$, so it lies below this instability boundary. Resilience changes the type and speed of adjustment while leaving the gain-one boundary unchanged for $\rho < 1$.

This is a local test with no further fundamental returns. A stable system can still exhibit several sales in succession and temporary amplification before settling. A fundamental innovation changes C, and reaching a new price peak activates the drawdown boundary. The calculation therefore does not establish a stationary distribution or predict the frequency, depth, or duration of noisy deleveraging episodes; those require the full dynamics studied in the simulations.

3.2 The Permanent-Impact Limit

At $\rho = 1$, $I_t = \lambda(B_t - B_0)$ and $W_t = C_t + \lambda B_0 = d_t + \lambda B_t$. At fixed W and zero fundamental increments, while the price stays below its running peak, the drawdown map is

$$d_{t+1} = W - \lambda L(d_t), \qquad \frac{\partial d_{t+1}}{\partial d_t} = \lambda |L'(d_t)|. \tag{3.9}$$

The equilibrium curve is $W = F(d) := d + \lambda L(d)$, with derivative $F'(d) = 1 - \lambda |L'(d)|$. For $G > 1$, it has a decreasing middle segment and two folds if both critical drawdowns are positive. The outer branches are stable on a fixed-W slice and the middle branch is unstable. This is the permanent benchmark's mechanism for different liquidation and recovery thresholds. For $\rho < 1$, those static folds are replaced by the single fixed point on each such fixed-C slice; they cannot be carried over as an O-W stability argument. Very slow resilience can nevertheless resemble the permanent benchmark over a finite observation interval.

3.3 Bounded Impact and Positive Fundamental Drift

Define $M_t = B_t - I_t/\lambda$. The impact recursion implies

$$M_{t+1} = \rho M_t + (1 - \rho) B_t. \tag{3.10}$$

Since $I_0 = 0$, $M_0 = B_0 \in [b_{\min}, b_{\max}]$. Both M_t and B_t remain in this interval, so $|I_t| \leq \lambda(b_{\max} - b_{\min})$, uniformly in ρ. Cumulative returns satisfy

$$p_t - p_0 = \sum_{s=1}^{t} r_c(s) = \sum_{s=1}^{t} \xi_s + I_t - I_0. \tag{3.11}$$

With iid Gaussian fundamentals of mean $\mu > 0$, the fundamental sum grows at rate μ almost surely, while displacement stays bounded. Prices make new records infinitely often. Each record has $d_t = 0$ and therefore implies $B_{t+1} = L(0) = b_{\max}$, irrespective of the next innovation. Exposure consequently returns to its ceiling repeatedly, without a deterministic recovery deadline under Gaussian noise.

If all subsequent fundamental increments instead equal the positive constant μ, the initial price peak is first regained no later than

$$T_{\text{peak}} \leq \left\lceil \frac{d_0 + \lambda(b_{\max} - b_{\min})}{\mu} \right\rceil, \qquad I_0 = 0. \tag{3.12}$$

This is a first-hit bound. Under transient impact, returning to full exposure can create positive displacement whose later decay induces another drawdown, even with positive constant fundamentals. Full exposure need not persist. At $\rho = 1$, this recovery reversal is absent.

Finally, impact recovery changes displacement, not cumulative fundamental returns. With zero future fundamental increments, a convergent trajectory on a fixed $C > 0$ slice has limiting exposure $B = L(C)$, which may fall short of the chosen recovery threshold. Local stability alone does not ensure convergence from a distant initial state. In the simulations we therefore distinguish a partial rebound, recovery to a near-normal exposure level, recovery to the original price peak, and a renewed selloff after recovery.

4 Daily Calibration and Deleveraging Times

4.1 Matching the Requested Exposure and Impact Levels

One period is one trading day. Exposure is now in the requested units: normal exposure is $b_{\max} = 0.5$ and the residual floor is $b_{\min} = 0.3$. The drawdown thresholds refer to cumulative simple-price losses from the running peak, rather than single-day returns. A smooth sigmoid approaches its endpoints asymptotically. We therefore interpret activation at a 5% price loss and completion at an 8% loss as 5% and 95% of the total 0.2 exposure reduction:

$$L(-\log 0.95) = 0.49, \qquad L(-\log 0.92) = 0.31, \qquad L(0) = 0.5. \tag{4.1}$$

Solving these conditions using the normalized sigmoid gives

$$\theta = 0.0673372052, \qquad w = 0.00544904452, \qquad A = 0.2000008594. \tag{4.2}$$

The sigmoid midpoint corresponds to a price loss of about 6.51%. Exposure declines continuously through this region and approaches 0.3 for larger drawdowns. Figure 1 shows the calibration.

An isolated sale of 0.05 must reduce the marked price by 30 basis points. Because impact is linear in log-price units, the exact coefficient and three-day surviving fraction are

$$\lambda = \frac{-\log(1-0.003)}{0.05} = 0.0600901804, \qquad h = 3, \qquad \rho = 2^{-1/3} = 0.793700526. \tag{4.3}$$

Thus $g(-0.05) = \log(0.997)$, whose simple-price effect is exactly -30 basis points. This fixes the immediate trade contribution; the full daily return also contains fundamentals and recovery of preceding displacement. The half-life is three trading days, not three calendar days.

Daily return volatility and mean are not determined by these four calibration targets. For the timing comparisons we retain the earlier scale illustration,

$$\sigma = 0.005, \qquad \mu = 0.0003125 = \sigma/16. \tag{4.4}$$

These are daily fundamental volatility of 0.5% and mean log return of 3.125 basis points. They are numerical assumptions, not estimates from data. The model formulation retains generic $\mu > 0$; sensitivity tests vary the mean and volatility independently.

The calibrated maximum gain is

$$G = \frac{\lambda A}{4w} = 0.551385 < 1. \tag{4.5}$$

Every zero-increment fixed point on a fixed positive-C slice is therefore locally stable for finite half-life. Feedback can prolong selling and produce damped fluctuations, but the calibration does not preserve the locally unstable cascade mechanism in the earlier high-impact examples. At the sigmoid midpoint, the eigenvalues are approximately $0.6725 \pm 0.3148i$, with modulus 0.7426. Holding the target fixed, the gain-one threshold requires about 54.34 basis points of immediate impact per sale of 0.05. The sensitivity study includes 60 basis points to cross that boundary without changing the requested baseline.

4.2 What the Timing Statistics Measure

An episode starts when exposure first reaches $B \leq 0.49$ while no episode is active. It ends at the first subsequent $B \geq 0.495$, within 1% of the normal level. This small difference between activation and recovery avoids counting every recrossing of 0.49 as a separate episode; it is a measurement convention, not a change in the trading rule. Reaching $B \leq 0.31$ identifies near-complete deleveraging. Many activated episodes recover before reaching that level, so their fraction is reported separately.

The reported times have different starting points:

- *Time without deleveraging*: from a recovery to the next activation. Small sigmoid-driven trades can still occur during this calm spell.
- *Interval between deleveragings*: from one activation to the next, including the intervening active episode and calm spell.
- *Duration of liquidation*: from activation to the first occurrence of that episode's minimum exposure.
- *Time to recovery*: from that minimum to the recovery threshold.
- *Total episode duration*: from activation to recovery, the sum of the liquidation and recovery phases.

The trough-based decomposition is retrospective; it does not assume traders know when the minimum will occur. Completed spells and episodes determine their mean durations, and unfinished observations remain censored. We also report waiting from a fresh price peak, $(B_0, d_0, I_0) = (0.5, 0, 0)$, to the first activation. That waiting time is not the same as a calm spell after an earlier recovery, which may end well below the previous price peak.

4.3 Simulation Design and Baseline Timings

The core sensitivity grid has 270 cells, crossing daily volatility $\sigma \in \{0.375\%, 0.5\%, 0.625\%\}$, daily mean $\mu \in \{1.5625, 3.125, 6.25\}$ basis points, impact $\{15, 22.5, 30, 45, 60\}$ basis points per sale of 0.05, and half-lives $\{1, 2, 3, 5, 10, \infty\}$ days. Eight additional specifications widen the volatility range or shift the drawdown band, two vary the recovery measurement threshold, and one sets impact to zero: 281 cases in total. Each uses 64 independent paths, 8,192 discarded days, and 65,536 retained days. The standard Gaussian innovations are shared across specifications. Independent validation uses 128 paths, 10,000 discarded days and 131,072 retained days, including the same two measurement sensitivities. Separate experiments use 4,096 matched paths per specification, from a fresh peak or an imposed 5% drawdown.

Table 2 gives the requested average times. An ordinary calm spell lasts 78.92 trading days. Activations occur 107.07 days apart, on average. An activated episode lasts 28.15 days: 13.25

days from activation to its first minimum exposure and 14.91 days from that minimum to recovery. The medians are much shorter than the means, reflecting a long right tail. Liquidation here is a phase containing possible interim purchases, not an uninterrupted sale run. Material sale runs, using $\Delta B \leq -0.0005$, average 1.90 days in the same natural sample.

Timing measure	Mean	95% interval	Median	90th pct.
Calm spell	78.92	[78.19,79.65]	10	260
Activation interval	107.07	[106.34,107.81]	32	312
Activation to trough	13.25	[13.04,13.45]	2	29
Trough to recovery	14.91	[14.69,15.12]	4	30
Whole episode	28.15	[27.77,28.54]	7	59
Fresh peak to first activation	225.50	[219.44,231.56]	162	490

Table 2: Baseline timing in trading days. The first five rows use independent natural validation: 128 paths, 10,000 discarded days, and 131,072 retained days per path. Episode phases use 156,354 completed episodes; 40 new episodes remain right-censored. Calm and activation intervals use completed intervals between observed events. The final row instead uses 4,096 paths initialized at a fresh peak; every first activation is observed within 20,000 days. Intervals are pointwise and account for clustering by independent path.

The system spends 74.95% of days at $B \geq 0.49$, and its average exposure is 0.4646. Exposure is at or below 0.31 on 11.82% of days. Thus this calibration, together with the assumed mean and volatility, does not preserve the earlier examples' roughly 99% normal-exposure occupancy. Starting from a fresh price peak gives a longer mean first waiting time, 225.50 days. By contrast, an ordinary calm spell often starts after only partial repair of the price drawdown. Figure 3 shows unconditioned paths and the episode decomposition.

4.4 Near-Complete Deleveraging Is Less Frequent and Lasts Longer

Only 18.25% of completed activated episodes reach $B \leq 0.31$ before recovering. The remaining episodes are partial reductions from 0.5. For the near-complete subset, it takes an average of 15.04 days to move from activation to $B \leq 0.31$, followed by 103.08 days to regain $B \geq 0.495$. The whole episode averages 118.12 days. Activations of episodes that reach 0.31 are about 584.79 days apart; partial episodes may occur between them. Table 3 distinguishes these deeper events from the averages across all activations.

Timing measure	Mean	95% interval	Median	90th pct.
Activation to $B \leq 0.31$	15.04	[14.90,15.18]	12	29
$B \leq 0.31$ to recovery	103.08	[101.30,104.87]	48	252
Whole near-complete episode	118.12	[116.35,119.89]	64	267
Interval between these activations	584.79	[576.72,592.86]	393	1373

Table 3: Trading-day timings for episodes reaching $B \leq 0.31$. Phase durations condition on the 28,529 completed near-complete episodes. The final row uses 28,423 observed intervals between activations of episodes known to reach 0.31; partial episodes can occur between them. Twenty-two additional near-complete episodes remain unrecovered at the horizon. A near-complete episode is selected by its realized depth and is not an unconditional forecast.

The independent first-passage experiment from a fresh peak reaches $B \leq 0.31$ after 662.39

days on average. That first-passage time can include several earlier partial episodes. It is different from both the duration of a single full reduction and the interval between successive deep episodes.

4.5 Sensitivity Around the Calibration

Table 4 and Figure 4 report nearby variations with fixed exposure bounds. Fundamental volatility and mean have the largest effects. At fixed mean 3.125 basis points, reducing daily volatility from 0.5% to 0.375% increases mean calm time from 79.4 to 222.1 days and raises occupancy above 0.49 from 74.8% to 91.0%. Increasing volatility to 0.625% reduces calm time to 45.8 days and occupancy to 59.5%. Matched paths in Figure 5 illustrate this change without selecting shocks or injecting stress.

Change from baseline	Calm	Interval	Liquidation	Recovery	Episode
Baseline	79.4	108.0	13.4	15.1	28.5
$\sigma = 0.375\%$	222.1	246.4	11.2	13.1	24.4
$\sigma = 0.625\%$	45.8	78.6	14.8	18.0	32.8
$\mu = 1.5625$ bps	53.6	108.9	23.9	31.3	55.2
$\mu = 6.25$ bps	189.4	203.7	6.4	7.9	14.3
Impact 15 bps	82.5	112.5	14.1	16.0	30.1
Impact 60 bps	69.3	93.0	11.3	12.4	23.7
$h = 1$ day	79.5	108.5	13.6	15.3	28.9
$h = 10$ days	80.9	109.5	13.5	15.1	28.6
Permanent impact	88.3	121.4	15.6	17.5	33.2
No impact	84.8	116.0	14.6	16.6	31.2
Drawdown 4–7%	54.1	82.3	13.3	14.9	28.2
Drawdown 6–9%	111.5	140.0	13.4	15.0	28.5

Table 4: Mean trading-day timings for one-at-a-time changes, using the 64-path matched grid. Liquidation means activation to trough; recovery means trough to $B \geq 0.495$. The mean and volatility vary independently. Impact is the simple-price effect of a 0.05 sale. All unspecified parameters remain at baseline, and all exposure bounds remain $[0.3, 0.5]$. The baseline row uses the grid seed, so it differs slightly from the larger independent validation in Table 2.

At fixed volatility 0.5%, halving the mean to 1.5625 basis points increases the mean active episode from 28.5 to 55.2 days. Doubling it to 6.25 basis points reduces the episode to 14.3 days and lengthens calm spells to 189.4 days. The onset interval need not move monotonically with drift over all values: a weaker mean shortens calm spells while lengthening active episodes. In the core grid, all 30 cases with volatility 0.375% and mean 6.25 basis points exceed 95% occupancy above 0.49; no other core combination of mean and volatility meets that occupancy threshold. This identifies the calm region on the tested slice, rather than an empirical calibration of those fundamental-return parameters.

Changing half-life between one and ten days has comparatively little effect near the baseline: mean active episodes remain around 28–29 days. Permanent impact lengthens the mean episode to 33.2 days and raises the fraction reaching 0.31 to 23.2%, compared with 18.2% at the requested three-day half-life. Increasing immediate impact from 15 to 60 basis points shortens mean calm spells from 82.5 to 69.3 days and raises the near-complete fraction from 16.2% to 27.4%, while shortening mean episodes from 30.1 to 23.7 days. Faster movement through the exposure range and subsequent impact recovery can therefore produce deeper but shorter threshold-defined episodes.

Shifting the response band from 5–8% to 4–7% lowers mean calm time to 54.1 days; shifting it to 6–9% raises calm time to 111.5 days. Mean episode duration stays near 28 days. A narrower 5.5–7.5% transition raises G to 0.827 and the near-complete fraction to 31.2%; a wider 4.5–8.5% transition lowers them to 0.413 and 12.2%.

The wider volatility checks show why the unspecified return distribution matters for waiting times. At daily volatility 0.25%, occupancy above 0.49 is 99.50%; at 1%, it is 30.92%, holding the positive mean fixed. In the fresh-peak experiment at 0.25%, 245 of 4,096 paths have not activated by day 20,000. The mean waiting time restricted at that horizon is 6,615.5 days; it must not be interpreted as an uncensored mean first-passage time.

4.6 Controls, Measurement Sensitivity, and Verification

From $(B_0, d_0, I_0) = (0.5, -\log 0.95, 0)$ with every future fundamental increment equal to its positive mean, minimum exposure is 0.4895 and recovery to 0.495 occurs at day 14. There are only two initial material sales. Starting at the sigmoid midpoint instead gives minimum exposure 0.3432 and recovery at day 65; starting at an 8% drawdown gives minimum 0.3014 and recovery at day 116. The corresponding zero-increment paths relax to their target exposures rather than fully recovering. These controls confirm moderate feedback at the requested gain, without an autonomous near-complete liquidation after every activation.

Matched Gaussian continuations from an imposed 5% drawdown yield a mean first episode of 21.46 days, and 12.30% reach 0.31 before recovery. The difference from natural episodes reflects the specified starting state, including zero inherited impact and no random overshoot of the activation threshold. Exposure recovery is also much earlier than recovery of the old price peak: the latter averages approximately 173 days from that imposed state. Reaching the exposure threshold does not mean the drawdown is zero.

Table 5 varies the recovery threshold on identical paths. This changes how nearby fluctuations are grouped into episodes, especially the recovery duration. The baseline level 0.495 is stated explicitly so the timing estimates can be reproduced or compared under another operational convention.

Recovery level	Calm	Interval	Liquidation	Recovery	Episode
0.49	56.2	75.0	8.9	9.8	18.8
0.495	78.9	107.1	13.2	14.9	28.2
0.499	132.3	187.4	25.0	30.1	55.1

Table 5: Sensitivity to the measurement convention, in trading days. These are identical baseline exposure paths; only the level ending an episode changes. Activation remains $B \leq 0.49$. Longer recovery requirements combine some nearby fluctuations into a single episode. The price and trading dynamics are unchanged.

Independent checks verify the exact target calibration, the 30-basis-point isolated trade, the three-day half-life, price and running-peak identities, exposure bounds, the permanent-impact limit, and episode accounting. The three studies use seeds 20261001, 20261002, and 20261003. Reported uncertainty is computed across independent paths, with ratio estimates for pooled event means. The means condition on completed spells unless a restricted first-passage mean is explicitly stated. The 36 episodes already active at the validation boundary are excluded from phase-duration averages; only the 156,354 completed new episodes enter them. The primary episode decomposition uses the first recorded minimum of B; resolving numerical floor ties through lagged drawdown shifts the baseline phase means by only 0.082 day and leaves total episode duration unchanged. The previous normalized results remain below as separate benchmarks.

5 Permanent-Impact Benchmark

This section preserves the previous numerical results with $b_{\max} = 1$, $b_{\min} = 0.2$, $\rho = 1$, and $I_0 = 0$. These are the earlier normalized comparisons, separate from the daily calibration in Section 4. The O-W comparison in Section 6 varies only impact persistence and uses new matched simulations.

5.1 Parameters and Operational Criteria

Scale return-like quantities by σ. At fixed impact persistence, the dimensionless parameters are μ/σ, θ/σ, w/σ, λ/σ, and the exposure floor. Multiplying $(\sigma, \mu, \theta, w, \lambda, d_0)$ by a common positive factor leaves the exposure dynamics unchanged. In the simulations, $\mu/\sigma = 1/16$ except in the zero-mean controls. The representative specification is

$$\frac{\theta}{\sigma} = 40, \quad \frac{w}{\sigma} = 1, \quad \frac{\lambda}{\sigma} = 10, \quad b_{\min} = 0.2. \tag{5.1}$$

For $\sigma = 0.005$, these values are $\mu = 0.0003125$, $\theta = 0.20$, $w = 0.005$, and $\lambda = 0.05$. The threshold is a 20% log drawdown, equivalent to an 18.1% price drawdown. This is a scale illustration, not an empirical calibration or a choice of calendar frequency.

The focused scan has 108 cells, holding the other parameters in (5.1) fixed:

$$\theta/\sigma \in \{20, 25, 30, 35, 40, 45, 50, 55, 60\},$$
$$\lambda/\sigma \in \{0, 2, 4, 5, 6, 8, 10, 15, 20, 25, 30, 40\}.$$

Each cell has 64 independent paths, 8,192 discarded periods, and 65,536 retained periods per path. Paths start at $(B, d) = (1, 0)$. The zero-impact column is a control. Common random innovations across cells improve comparisons; the independent paths determine standard errors. A separate comparison checks the effect of removing partial adjustment. The heatmap describes a slice of parameter space, not an exhaustive classification of every target width, drift, and floor.

We classify a cell as exhibiting the requested behavior when all of the following hold:

1. The simulated fraction of periods with $B \geq 0.95$ is at least 95%.
2. Starting from $B_0 = 1$ and $d_0 = \theta - 2\sigma$, with all future fundamental increments fixed at μ, exposure falls to 0.6 or below.
3. The minimum exposure is at least 0.2 lower than under the matched zero-impact experiment, and there are at least five consecutive material sales, where a material sale means $\Delta B_t \leq -0.001$.
4. After this stressed episode, exposure recovers to $B \geq 0.95$ within 5,000 periods under the positive fundamental increments.

These are explicit numerical criteria for "usually calm," "amplified deleveraging," and "recovery." They are not universal economic thresholds. The controlled initial state has $I_0 = 0$ and represents an accumulated drawdown; it is not treated as a single Gaussian shock. Unforced Gaussian simulations separately check that selloffs actually occur without injecting such an initial state. The pointwise occupancy intervals use independent-path standard errors; cells whose intervals straddle 95% are marked as uncertain.

5.2 Is Partial Adjustment Necessary?

The mechanism survives full adjustment at the previous impact value $\lambda/\sigma = 20$. The matched comparison below holds all parameters and Gaussian innovations fixed except a, using 128 paths with 10,000 discarded and 65,536 retained periods. These comparison data were generated in the previous speed study; the new baseline is validated independently below.

Rule	Time $B \geq 0.95$	Stress minimum B	Sales in run	Recovery period
Partial, $a = 0.30$	98.33%	0.200008	19	186
Full, $a = 1$	98.34%	0.200001	4	174

Table 6: Removing partial adjustment at fixed $\lambda/\sigma = 20$, $\theta/\sigma = 40$, $w/\sigma = 1$, and $b_{\min} = 0.2$. Occupancy uses natural Gaussian paths; the other columns use the controlled stress with constant positive fundamentals.

Full adjustment preserves high normal exposure, amplified liquidation, and recovery. It compresses the selloff, so the four-sale stress at impact 20 fails the existing five-sale operational criterion. This is a duration restriction, not the disappearance of feedback. We retain that criterion and choose $\lambda/\sigma = 10$ for the simplified representative case: there are six material sales, minimum exposure is 0.202962, and recovery occurs at period 59. The adjustment fraction is fixed at one throughout the new scan and is no longer a free model parameter. This change of the representative impact value is separate from the fixed-parameter comparison above.

5.3 The Region Found in the Scan

With full adjustment, 14 of the 108 tested cells meet all four criteria. In every one of these cells the lower endpoint of the pointwise 95% occupancy interval also exceeds 95%. The exact tested combinations are summarized in Table 7 and outlined in Figure 6.

Threshold θ/σ	Working impact values λ/σ	Qualification
20, 25	None	Calmness criterion fails for sustained cascades
30, 35, 40, 45, 50, 55, 60	8, 10	All pass with occupancy margin

Table 7: Working cells under full adjustment and the unchanged five-sale persistence criterion. An occupancy margin means the pointwise 95% lower confidence endpoint exceeds 95%; this is not a simultaneous confidence guarantee for the selected region.

With $w/\sigma = 1$ and $b_{\min} = 0.2$, the analytical gain threshold is approximately $\lambda/\sigma > 5$. In the prescribed controlled stress, tested impacts at or below 6 do not cause deep deleveraging. Impacts 8 and 10 produce eight and six consecutive material sales, respectively. Tested impacts from 15 through 40 still generate deep liquidation and recovery, but the liquidation takes only three or four material sales and fails the five-sale criterion. The narrow outlined impact range therefore reflects our requested duration as well as the underlying feedback.

At $\theta/\sigma = 40$, near-normal occupancy is 98.90% at impact 8 and 98.83% at impact 10 in this grid. The latter cell has a 95% interval [98.63%, 99.02%]. These are finite-grid results, not exact continuous boundaries. The two occupancy intervals that straddle 95%, at $(\theta/\sigma, \lambda/\sigma) = (30, 15)$ and $(35, 30)$, belong to cells that already fail the five-sale requirement.

5.4 Independent Natural Simulations

The representative parameters were checked with a separate seed using 128 independent paths, 10,000 discarded periods, and 65,536 retained periods per path. A deleveraging episode begins at the first crossing of $B \le 0.6$ while no episode is active and ends at the first subsequent $B \ge 0.95$. Episodes already active at the end of burn-in are excluded from duration summaries. New episodes still active at the horizon are reported as censored, not as permanent failures to recover.

	Candidate	Zero impact	Zero mean
Time with $B \ge 0.95$	98.89%	99.09%	16.00%
95% interval	[98.74,99.03]	[98.96,99.22]	[13.50,18.49]
New episodes	1,010	1,054	3,176
Recovered by horizon	1,008	1,052	3,079
Right-censored episodes	2	2	97
Median completed duration	39	22	62
90th percentile completed duration	233	150	1,301

Table 8: Independent natural simulations of the permanent-impact benchmark. Durations are periods from the first $B \le 0.6$ crossing to recovery. Duration quantiles condition on completion. The zero-mean column is a finite-horizon counterfactual; it is not presented as a stationary distribution.

The candidate is near normal exposure about 98.9% of the time and still produces hundreds of spontaneous deleveraging episodes across the independent paths. In the candidate, a material sale is followed by another material sale 47.4% of the time, versus 43.2% without impact. This event-oriented measure describes persistence of selling; it is not an assertion about the sign of the unconditional stationary autocorrelation function. The controlled experiments below establish the feedback mechanism more directly than trade persistence alone, and the same controls show whether impact sustains those trades.

5.5 Controlled Cascades and Recovery

Initialize $(B_0, d_0) = (1, 38\sigma)$ and set every subsequent fundamental increment to its positive mean $\mu = \sigma/16$. The first four sales have magnitudes 0.0954, 0.1031, 0.1735, and 0.2859; the next two are 0.1317 and 0.0074. Sales initially accelerate even though every new fundamental return is positive. There are six consecutive material sales, minimum exposure is 0.202962, and exposure recovers above 0.95 at period 59. Without impact, only the first trade is a sale, leaving exposure at 0.904638; positive fundamentals then raise the target and induce purchases. Thus the subsequent liquidation is generated by impact changing the target, without partial adjustment or negative fundamental increments.

For the candidate $G = 2$, which is also the maximum local multiplier. In units of σ, the high-exposure fold is at $(d/\sigma, B, W/\sigma) = (38.237, 0.8828, 47.066)$ and the low-exposure fold is at $(41.763, 0.3172, 44.934)$. The initial $W_0/\sigma = 48$ lies beyond the high-exposure fold. Positive fundamental increments lower W toward recovery. The conservative bound (3.12) for complete price recovery is 736 periods; exposure recovery at period 59 occurs earlier and does not mean the old price peak has already been regained.

If all future fundamental increments are instead exactly zero, the stressed system converges near $B = 0.202018$ and does not recover. This counterfactual removes both drift and random fluctuations. Under zero-mean Gaussian noise, favorable shocks can still generate recovery.

Matched stochastic continuations reinforce the deterministic evidence. Across 4,096 pairs starting at $(1, 38\sigma)$, impact lowers mean minimum exposure over the first 100 periods by 0.192 (95% paired interval $[0.185, 0.199]$ for the additional decline). The fraction with at least five consecutive material sales in the first 200 periods rises from 51.1% without impact to 70.4% with impact, although the median longest run is five in both cases. For a separate recovery experiment, both drift specifications start from the same depleted state, $(B_0, d_0/\sigma) \simeq (0.200269, 47.9973)$, and use identical Gaussian innovations. Recovery within 1,000 periods occurs in 99.24% of positive-mean paths and 82.89% of zero-mean paths; within 5,000 periods the fractions are 100% and 91.99%. These are finite-sample fractions, not deadline guarantees.

6 Comparison with O-W-Style Resilience

6.1 Matched Design and Diagnostic Measures

The comparison crosses the same 108 threshold–impact cells with

$$h \in \{0, \tfrac{1}{2}, 1, 2, 4, 8, 16, 40, 80, \infty\}.$$

All 1,080 cells retain the earlier normalization $b_{\max} = 1$ and use $\sigma = 1$, $\mu = 1/16$, $w = 1$, $b_{\min} = 0.2$, 64 independent paths, 8,192 discarded periods, and 65,536 retained periods. Initial conditions are $(B_0, d_0, I_0) = (1, 0, 0)$, with common standard Gaussian innovations across parameter cells. These are new matched draws, so the permanent column is a replication of Section 5, rather than a reuse of its sample. Figure 2 first isolates the mechanical impact recovery from the endogenous response.

The four operational criteria remain unchanged. For each cell, the 5,000-period controlled stress starts at $(1, \theta-2, 0)$ and compares constant fundamental increments $1/16$ with zero impact and with zero fundamental increments. We also report minimum exposure, additional exposure decline, initial and longest material-sale runs, first recovery, and subsequent selloffs separately. The distinction between initial and longest sale runs is essential: a later fluctuation can satisfy the original five-sale condition.

The matched stress study uses 4,096 independent Gaussian paths over 5,000 periods at $\theta = 40$, $\lambda \in \{10, 20\}$, and every half-life, starting at $(1, 38, 0)$. The same standard innovations are used for positive-mean and zero-mean continuations and no-impact controls. A separate recovery study starts every impact specification from the same depleted state for its λ: the state reached after 40 permanent-impact periods with zero fundamental increments. This sensitivity deliberately inherits a negative I_0 from that history; it is the sole exception to a fresh initialization with $I_0 = 0$.

An episode begins at $B \le 0.6$ and ends at the next $B \ge 0.95$. Duration summaries condition on completion; unfinished episodes remain censored. A renewed crisis is a new episode within 100 periods after recovery. The denominator includes only recoveries with the full 100-period follow-up: all eligible recoveries in the natural study, and eligible first recoveries in the stress study. The saved stress results also report 500-period renewal. These event measures do not assume a stationary autocorrelation structure.

6.2 The Region Depends on Which Form of Feedback Is Required

Table 9 reports the full-grid classification, and Figures 9–10 display the components and the tested parameter region. The original criteria admit more cells with rapid resilience, but this is largely a change in the form of trading. Every qualifying cell with $h \le 1$ obtains its five consecutive material sales in a later run. None has five sales in its initial liquidation. Thus the larger outlined region is not evidence of a broader initial-cascade region. Strong impact can empty most of the position in fewer periods and then generate repeated purchases and sales.

h	Original tests	Lower-CI test	Later-run only	Initial cascade
0	29	27	29	0
0.5	30	28	30	0
1	39	35	39	0
2	47	42	39	8
4	24	21	16	8
8	16	14	0	16
16	16	14	0	16
40	16	14	0	16
80	14	14	0	14
∞	14	14	0	14

Table 9: Qualifying cells out of 108 at each half-life. Original tests retain all four benchmark criteria. Lower-CI test additionally requires the pointwise 95% occupancy interval to lie above 95%. Later-run only counts qualifying cells whose first sale run has fewer than five material sales; the final column removes them as a diagnostic, without changing the original classification.

Using the initial sale run as an additional diagnostic gives a simple description of the tested region. At $h = 2$ or 4, the initial-cascade cases have $\lambda/\sigma = 10$ and $\theta/\sigma \ge 25$. At $h = 8, 16$, or 40, they have $\lambda/\sigma \in \{8, 10\}$ and $\theta/\sigma \ge 25$. At $h = 80$ or permanent impact, the corresponding threshold is $\theta/\sigma \ge 30$. These are statements about the listed grid points, not continuous boundaries. The qualifying threshold-25 cases have occupancy intervals crossing 95%; at thresholds 30 and above the corresponding lower bounds exceed 95%. No threshold-20 case meets all four original criteria.

The analytical results explain part of this pattern. Here $G \simeq \lambda/5$, so impact above five permits a local gain above one. This is a necessary description of an amplifying part of the target, not a sufficient classifier of the stressed path. At the zero-increment fixed point with $C = 38$, $k \simeq 0.840$ for $\lambda = 10$ and $k \simeq 1.680$ for $\lambda = 20$. The former fixed point is locally stable for every finite half-life; the latter is unstable. During a large displacement, however, $d = C - I$ can traverse the steeper part of the target, so even the locally stable case can undergo a substantial initial liquidation. Resilience changes that excursion and the subsequent oscillations without changing the local gain-one boundary.

6.3 Moderate Resilience Preserves the Cascade and Speeds Recovery

At the representative $\theta = 40$, $\lambda = 10$, constant-positive fundamentals produce a deep selloff at every tested $h \ge 2$. At $h = 2$, there are five initial material sales, minimum exposure is 0.4687, and the first recovery occurs at period 8. At $h = 4$, there are six initial sales, minimum exposure is 0.2275, and recovery occurs at period 11. At $h = 8$, the corresponding minimum and recovery period are 0.2094 and 13. Permanent impact gives six sales, minimum 0.2030, and recovery at period 59. With $h = 0, \frac{1}{2}$, or 1, the prescribed stress never reaches $B \leq 0.6$: impact recovery cushions this particular initial liquidation. It does not eliminate deep episodes driven by Gaussian shocks.

At $\lambda = 20$, the controlled system reaches near-floor exposure for every half-life. The initial run contains four material sales throughout. Recovery occurs at period 6 under complete resilience, versus period 174 with permanent impact. In the former case another deep episode begins at period 11, despite every fundamental increment being positive. Figure 11 shows how a rapid rebound can reverse again.

An independent validation uses 128 paths, 10,000 discarded periods and 65,536 retained periods, with a new seed. It repeats every qualifying finite-half-life baseline case at $\theta = 40$, $\lambda = 10$, together with the permanent case and the same half-lives at $\lambda = 20$: 14 cases in total. All seven baseline cases confirm high normal-exposure occupancy. At $h = 4$, occupancy is 99.15%, compared with 98.85% under permanent impact; the paired difference is 0.301 percentage points, with a pointwise 95% interval $[0.267, 0.335]$. Table 10 also shows the tradeoff: the median completed episode is shorter, but renewed selloffs are more common. Figure 12 shows fixed unconditioned paths, including paths without deep episodes in the displayed window.

h	High exposure (%)	95% interval	Episodes	Median	Renewed (%)
2	99.14	[99.01,99.27]	2905	7	80.9
4	99.15	[99.02,99.28]	2429	10	76.7
8	99.16	[99.03,99.29]	1960	14	70.1
16	99.16	[99.04,99.29]	1612	18	63.0
40	99.14	[99.02,99.27]	1309	25	53.9
80	99.11	[98.99,99.24]	1169	30	48.8
∞	98.85	[98.70,99.01]	963	41	39.6

Table 10: Independent natural validation at $\theta/\sigma = 40$ and $\lambda/\sigma = 10$: 128 paths, 10,000 discarded periods and 65,536 retained periods. Median is completed episode duration. Renewed is a new deep episode within 100 periods of a recovery with full follow-up. All new finite-h episodes in this table complete; one permanent-impact episode remains right-censored. Intervals describe occupancy, not the conditional duration or recurrence statistics.

6.4 Faster Individual Recoveries Can Mean More Repeated Selloffs

The matched Gaussian stress experiments corroborate the cushioning effect while separating it from recurrence (Table 11 and Figure 13). Relative to permanent impact, complete resilience raises mean minimum exposure over the first 100 periods by 0.0761 at $\lambda = 10$ and 0.0456 at $\lambda = 20$. The respective paired 95% intervals are $[0.0712, 0.0810]$ and $[0.0410, 0.0503]$. These comparisons use all paths, so they do not condition on which paths recover.

At $\lambda = 20$, the median completed first episode lasts 3 periods with complete resilience and 117 with permanent impact. But a renewed deep episode within 100 periods of the first recovery occurs in 83.3% versus 11.0% of eligible recovered paths. Across all 4,096 paths and 5,000 periods, there are 57,428 versus 6,578 episodes. At $\lambda = 10$, the analogous renewal fractions are 83.0% and 38.0%. These conditional fractions are descriptive: eligibility differs between specifications. They should be read alongside the all-path depth, occupancy, and episode counts.

Turnover rises as well. Complete resilience increases gross exposure turnover over the first 200 periods by a paired mean 5.666 at $\lambda = 10$ (95% interval $[5.462, 5.870]$) and 9.799 at $\lambda = 20$

λ/σ	h	Min. B	Initial sales	Duration	Renewed (%)	High (%)
10	0	0.362	2.61	4	83.0	96.71
10	2	0.325	3.25	6	75.3	96.83
10	8	0.301	3.58	11	63.5	96.90
10	∞	0.286	3.73	40	38.0	95.78
20	0	0.261	3.18	3	83.3	96.80
20	2	0.233	3.56	7	73.6	96.92
20	8	0.220	3.58	15	57.4	96.91
20	∞	0.215	3.58	117	11.0	93.85

Table 11: Matched positive-mean Gaussian stress experiments, $\xi_t \sim N(\sigma/16, \sigma^2)$: 4,096 paths from $(B_0, d_0/\sigma, I_0) = (1, 38, 0)$ over 5,000 periods. Min. B is the mean path minimum in the first 100 periods; initial sales is the mean first material-sale run; duration is the median completed first-episode duration. Renewed uses eligible first recoveries with 100 periods of follow-up. High is full-horizon occupancy above 95% exposure, conditional on the stressed start. Different eligible recovery subsets mean the duration and recurrence columns are descriptive, not paired treatment effects.

([9.506, 10.092]). The mechanism is symmetric: decaying negative displacement supports purchases, then decaying positive displacement can induce another drawdown. A shorter individual crisis therefore does not imply fewer crises or less trading.

6.5 What Positive Drift Adds, and What Impact Recovery Cannot Do

With zero subsequent fundamental increments, resilience can reverse the impact component of a loss without repairing its fundamental component. For $\lambda = 10$, $h = 2$, and $(B_0, d_0, I_0) = (1, 38, 0)$, exposure rebounds above 0.95 at period 9, then converges to $L(38) = 0.9046$. It does not regain the original price peak. For $\lambda = 20$ at the same start, every tested finite half-life produces large repeated fluctuations throughout the 5,000-period zero-increment run; permanent impact remains depleted. The final-window trajectories in Figure 11 document persistence over this horizon, rather than proving a limit cycle. Starting at $d_0 = 40$ instead gives persistent finite-horizon oscillations at both impact strengths for every tested finite half-life.

The depleted-state Gaussian comparison makes the distinction between a first rebound and normal exposure especially clear. At $\lambda = 20$, $h = 0$, all positive-mean and zero-mean paths cross $B \geq 0.95$ within 1,000 periods. Under permanent impact, the fractions are 98.68% and 70.68%, respectively; by 5,000 periods they are 100% and 86.89%. Yet for transient impact the last-1,000-period high-exposure occupancy is about 99% with positive mean and only 37–39% with zero mean. Those are conditional finite-horizon occupancies, not stationary-distribution claims. Positive drift supplies the long-run price growth established in Section 3; recovery of impact alone can provide a short-lived exposure rebound.

6.6 Verification and Reproducibility

The numerical implementation reproduces the permanent-impact recursion at $\rho = 1$ to floating-point accuracy. Independent reconstructions verify geometric decay after an isolated trade, the price identity (3.11), drawdown from the running peak, and exposure and impact bounds. Analytic eigenvalues were checked against finite-difference Jacobians in 240 cases. Episode entries, completions, censoring, and renewal denominators reconcile, including episodes crossing the burn-in boundary.

The natural grid, validation, and stress studies use seeds 20260925, 20260926, and 20260927, respectively. Saved outputs contain every parameter cell, each operational criterion, per-path statistics, paired comparisons, episode records, and plotting traces. Occupancy intervals use independent path fractions; paired intervals use path-level differences. All intervals are pointwise, with no multiple-comparison correction. Neither discarded initial periods nor a finite simulation establishes stochastic stationarity.

7 Interpretation and Scope

The requested daily calibration constrains the mechanism more tightly than the earlier normalized experiments. A 30-basis-point impact for a 0.05 sale and a smooth 5–8% drawdown transition imply maximum gain 0.55. The system can sell repeatedly as losses accumulate, but it does not have the locally unstable feedback of the earlier gain-above-one examples. This distinction is a consequence of the calibration, rather than a numerical failure to find the earlier behavior.

Under the assumed daily fundamental distribution, most activations are partial. Average calm spells, start-to-start intervals, liquidation phases, and recovery phases answer different questions. A near-complete reduction to 0.31 is much less frequent and takes much longer to recover from than the average activated episode. Recovery to 0.495 also need not restore the old price peak. Stating event thresholds and conditioning is therefore essential to interpreting the reported times.

The assumed mean and volatility determine whether normal exposure is common or nearly continuous. The baseline spends about 75% of days above 0.49; nearby lower-volatility, higher-mean specifications can exceed 95%. At this moderate feedback gain, changing impact's half-life around three days affects timing much less than changing fundamentals. Stronger impact can deepen episodes while shortening their duration through faster movement and subsequent recovery of displacement.

All timing estimates are finite-horizon simulation results under Gaussian fundamentals, not empirically fitted predictions. The formulation retains generic $\mu > 0$, immediate target adjustment, and no recovery reset. Positive drift guarantees repeated visits to the exposure ceiling but does not exclude future deleveraging. Earlier permanent and transient high-impact comparisons remain as distinct benchmarks; the archived rolling-window and partial-adjustment studies are unchanged.

With the factor interpretation in Section 1, all reported drawdowns and episode times refer to the systematic crowding index. The scalar innovation ξ_t has the same law and enters the same recursion as in the simulations; this notation change does not alter their numerical results. The simulations do not include the additional residual contributions to fund P&L in (1.4).

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Anna A. Obizhaeva and Jiang Wang. Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets*, 16(1):1–32, 2013. doi: 10.1016/j.finmar.2012.09.001.

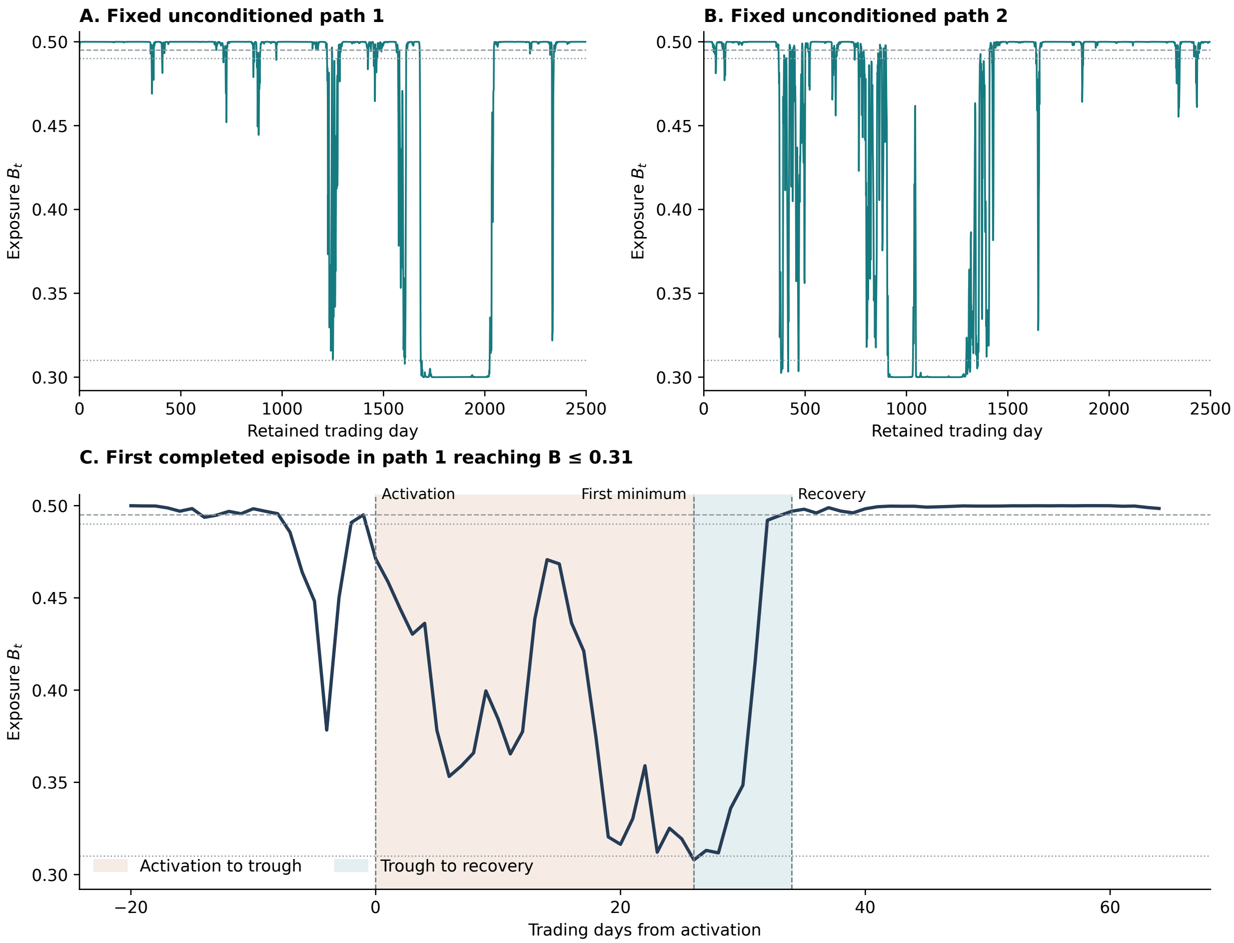

Figure 3: Calibrated daily exposure with $\sigma = 0.5\%$, mean 3.125 basis points, impact 30 basis points per 0.05 sale, and half-life three days. Upper panels show the first two fixed validation paths over their first 2,500 retained trading days. Lower panel selects the first completed episode in path 1 that reaches $B \leq 0.31$, to illustrate activation, the first minimum, and recovery. This lower panel conditions on a deep event; it is not a typical-path forecast. Near-complete deleveraging and recovery thresholds are 0.31 and 0.495, respectively.

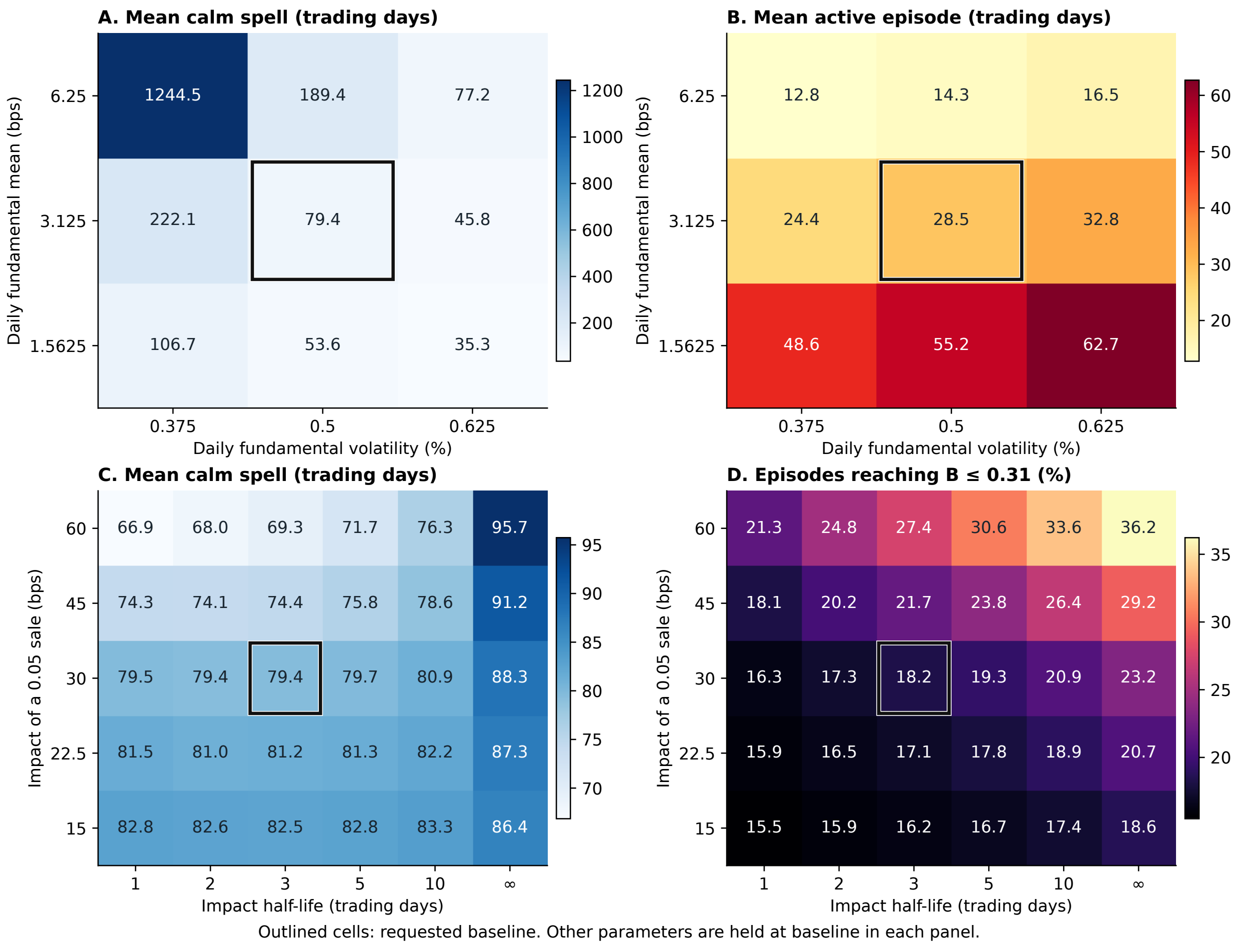

Figure 4: Sensitivity near the daily calibration. Upper panels vary fundamental mean and volatility independently, at baseline impact and half-life. Lower panels vary impact and half-life, at baseline mean and volatility. Black outlines identify the requested baseline. Calm time is recovery to next activation; episode duration is activation to recovery. Panel D conditions on completed episodes. Cells show discrete simulated values and are not interpolated parameter boundaries.

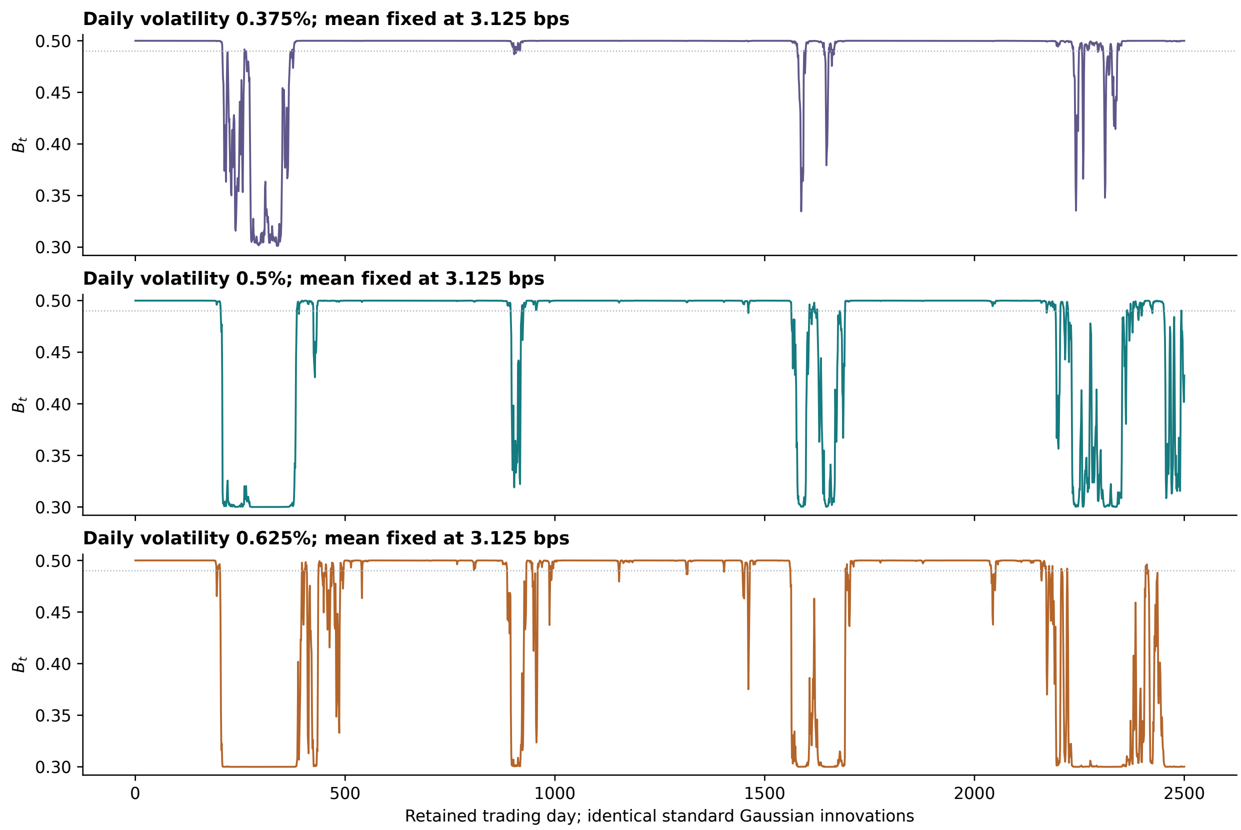

Figure 5: Matched volatility sensitivities, using the first fixed grid path and identical standard Gaussian innovations across specifications. The mean remains 3.125 basis points per day; drawdown band, exposure bounds, impact and half-life remain fixed. Every panel displays the same first 2,500 retained trading days. Increasing volatility changes the frequency and persistence of drawdowns even though impact's local feedback gain is unchanged.

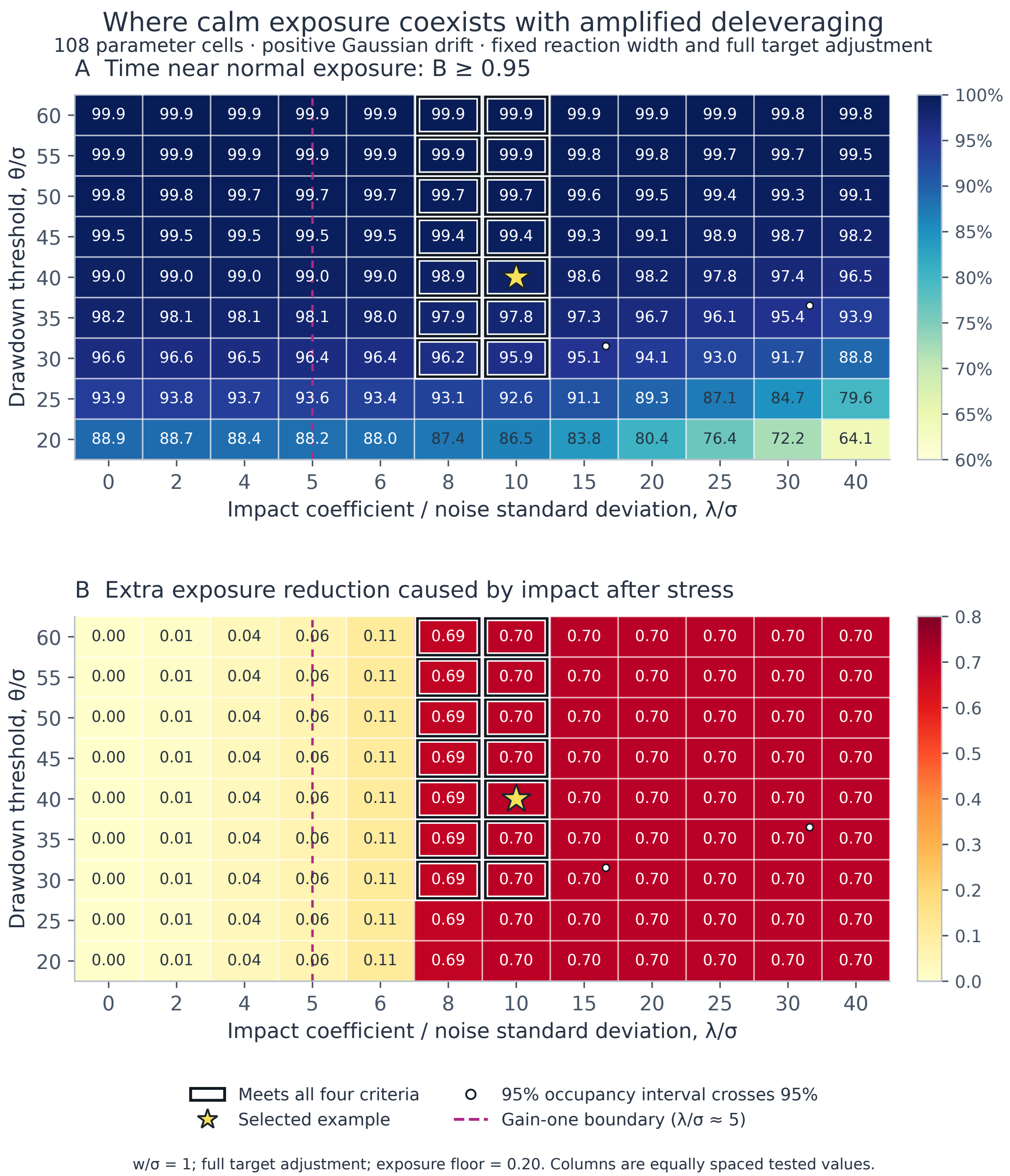

Figure 6: Permanent-impact benchmark ($\rho = 1$): parameter region with $w/\sigma = 1$, $b_{\min} = 0.2$, and full adjustment to the target each period. Upper panel: the fraction of retained periods with $B \geq 0.95$. Lower panel: the additional exposure decline caused by impact in the controlled stress. Outlines mark cells meeting all four operational criteria; uncertainty markers identify occupancy intervals containing 95%. Columns are the tested impact coefficients, with equal visual spacing. The dashed boundary is the analytical onset of gain above one, approximately $\lambda/\sigma = 5$. Some unoutlined high-impact cells still crash and recover but liquidate in fewer than five material sales.

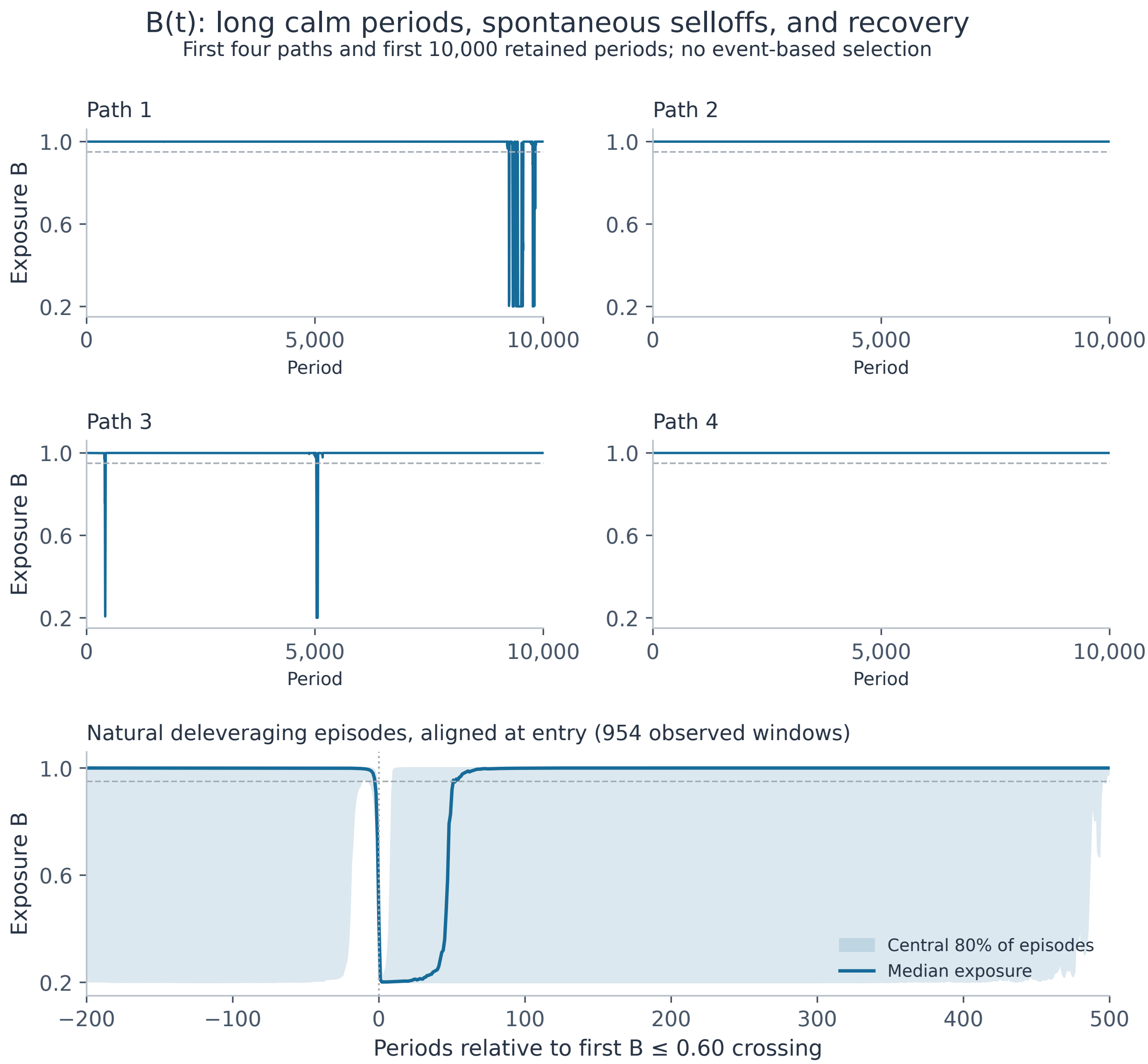

Figure 7: Permanent-impact benchmark ($\rho = 1$): Gaussian simulations of normalized exposure under (5.1), with no injected stress or manual reset. The upper panels show fixed, unconditioned paths. The lower panel aligns natural episodes at their first crossing of $B \leq 0.6$ and shows the median and central 80% range. Alignment conditions on an event, so this panel is not an unconditional forecast.

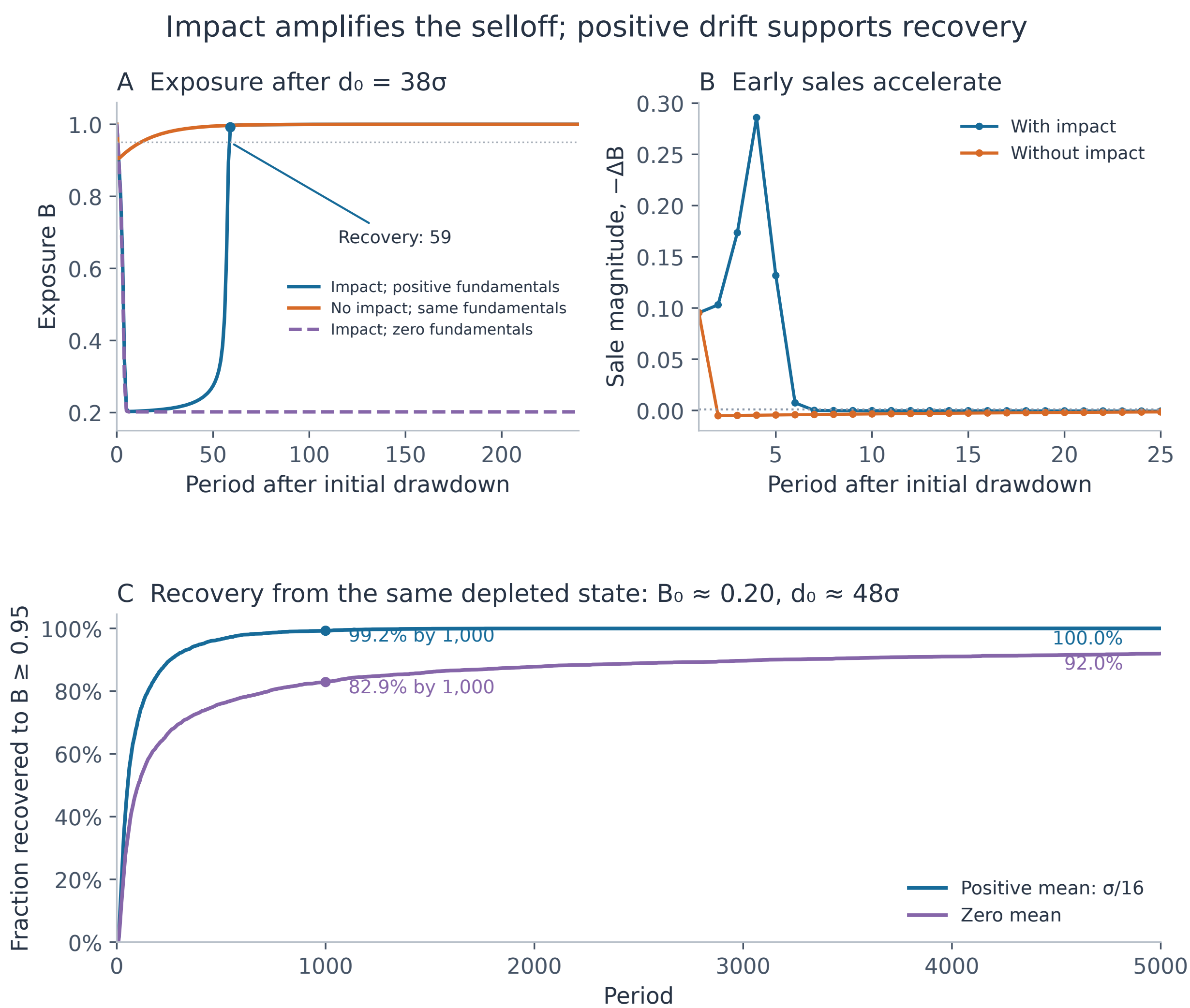

Figure 8: Permanent-impact benchmark ($\rho = 1$): the controlled drawdown experiment starts from $B_0 = 1$ and $d_0 = 38\sigma$. Upper panels: exposure and the first 25 trades under positive constant fundamentals, with and without impact; the third exposure path keeps impact but sets every subsequent fundamental increment to zero. Lower panel: stochastic recovery probabilities from the same depleted initial state, under matched Gaussian innovations with positive or zero mean. A recovery means at least one subsequent crossing of $B \geq 0.95$; exposure can fall again later.

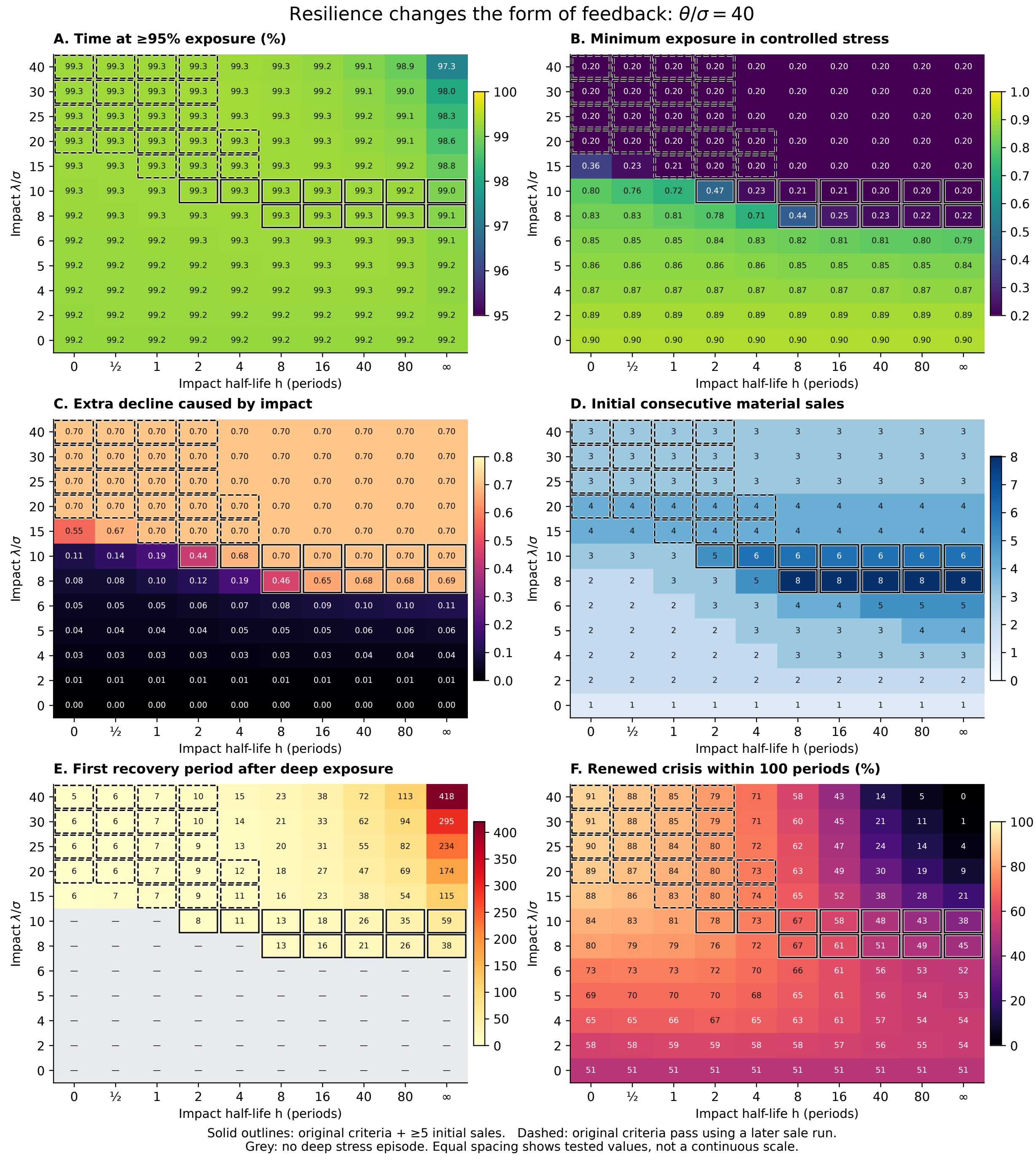

Figure 9: The resilience–impact slice at $\theta/\sigma = 40$. Panels A and F use natural Gaussian paths; B–E use the constant-positive controlled stress from $(1, 38, 0)$. Outlines retain the original four operational criteria: solid outlines also have five initial material sales; dashed outlines need a later sale run. Panel D displays the initial run to reveal this distinction. A dash in panel E means no deep episode occurred. Renewal in F is conditional on recoveries with 100 observed subsequent periods. All tested values have equal visual spacing; the figure does not interpolate between cells.

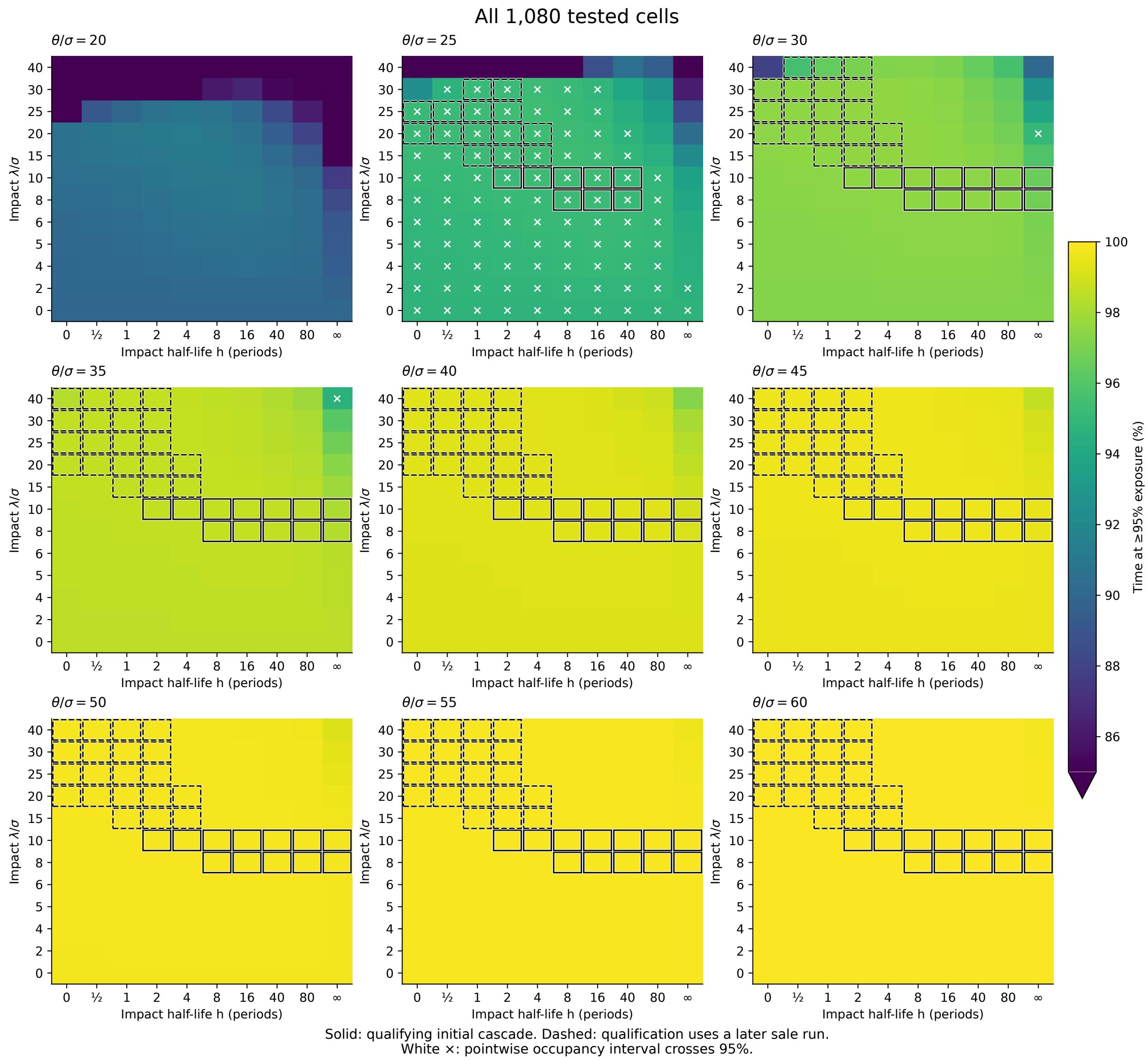

Figure 10: All 1,080 threshold–impact–resilience cells. Color is the natural fraction of time above 95% exposure. Solid outlines pass the original four criteria and have five initial consecutive material sales; dashed outlines pass using a later run. White crosses mark pointwise occupancy intervals containing 95%. The lower color endpoint clips occupancy below 85%. Table 9 gives counts and a diagnostic based on the first sale run. Parameter axes show discrete tested values.

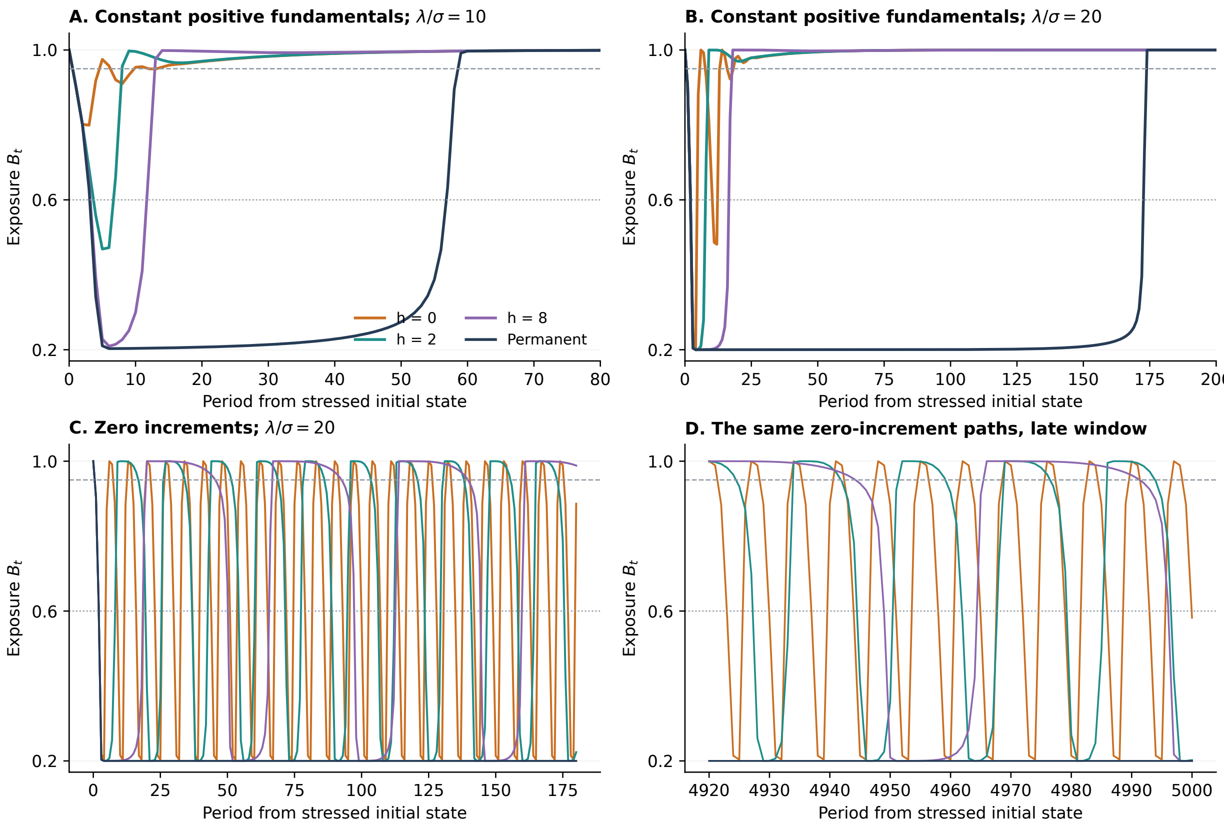

Figure 11: Controlled exposure dynamics from $(B_0, d_0/\sigma, I_0) = (1, 38, 0)$. Upper panels: every fundamental increment equals $\mu = \sigma/16$. At $\lambda/\sigma = 10$, complete resilience prevents a deep initial episode; at 20, it causes a rapid recovery and another selloff. Lower panels: all fundamental increments are zero, with impact strength 20. Large oscillations persist into the final 81 periods of the 5,000-period run under transient impact, while permanent impact stays depleted. This is a finite-horizon observation, not a proof of a periodic orbit. Dashed and dotted lines mark $B = 0.95$ and $B = 0.6$. Colors identify the same half-lives in all panels.

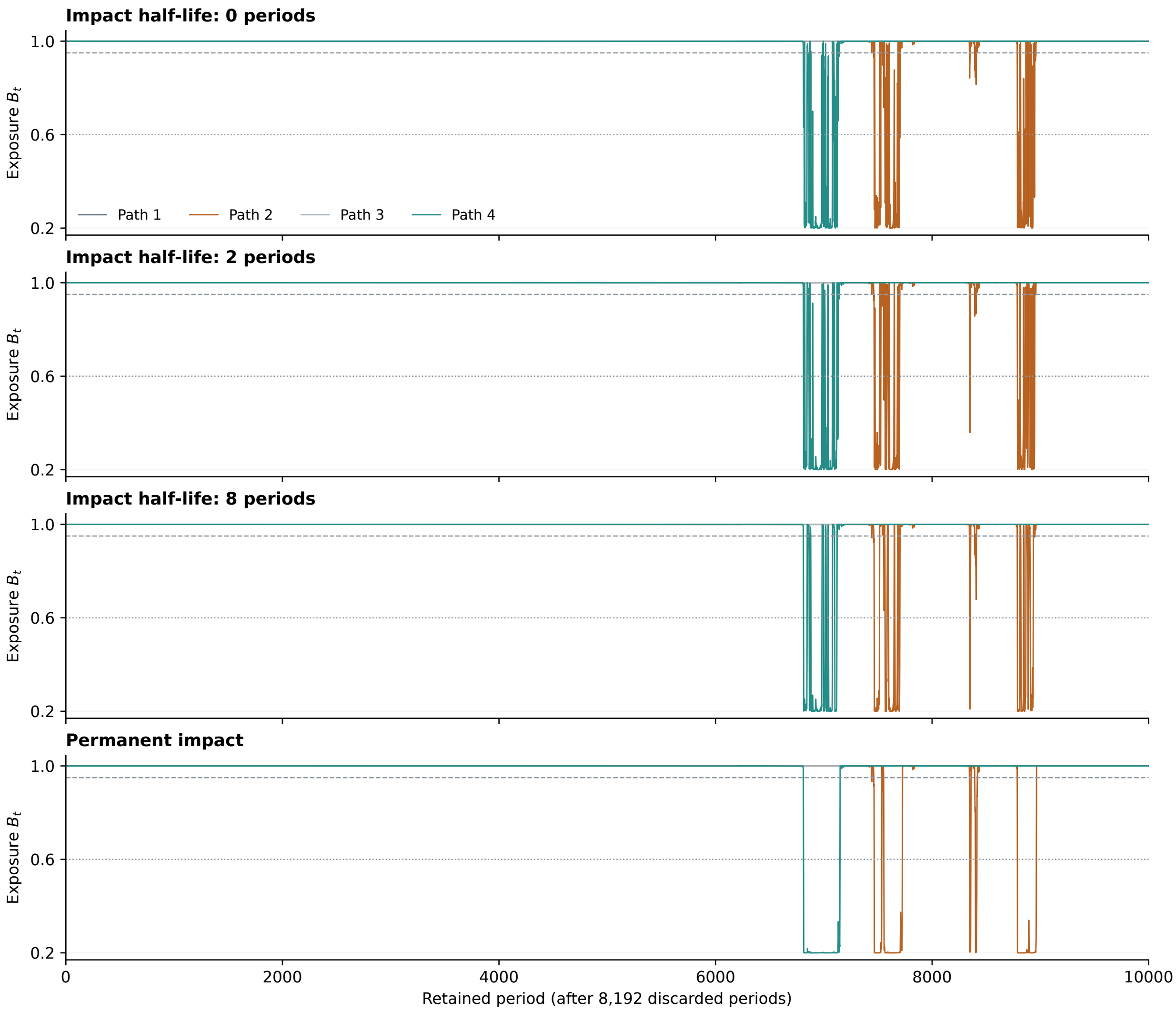

Figure 12: Matched natural Gaussian exposure paths at $\theta/\sigma = 40$, $\lambda/\sigma = 10$, and $\mu/\sigma = 1/16$. The first four independent chains and first 10,000 retained periods are displayed without selecting events; their innovations match across half-lives. Paths 1 and 3 have no deep episode in this window. Resilience shortens episodes while allowing more repeated entries. The full statistics use 64 chains and 65,536 retained periods per cell, with independent validation in Table 10.

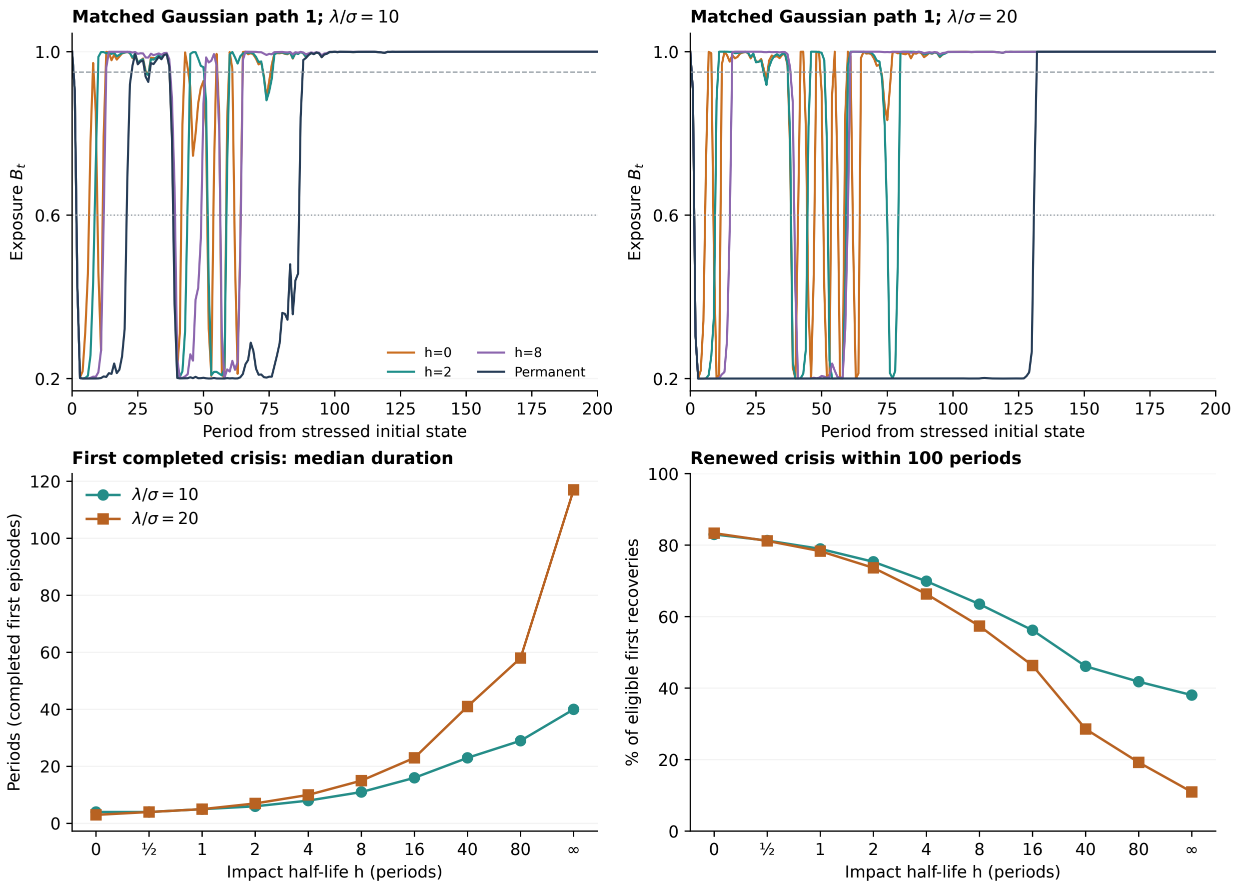

Figure 13: Matched Gaussian stress experiments from $(1, 38, 0)$, with positive mean $\mu/\sigma = 1/16$. Upper panels show the first fixed path and first 200 periods, using identical innovations across impact specifications. Lower panels summarize all 4,096 paths over 5,000 periods. Faster resilience shortens the median completed first episode but raises the frequency of renewed deep exposure after the first recovery. The renewal denominator includes only first recoveries with 100 periods of follow-up; duration and recurrence therefore condition on specification-dependent subsets.